Full-corpus results: what we tested and what we found

Run: 2026-11-06_01-23-40-040fd7b7. Analysis: full-corpus-report-v2.

Interactive tables · Case explorer

Continuous PDF · Detailed setup guide (Spanish)

Corpus correction and response replacement

This is the corrected principal corpus, composed from two executions. The original run supplied 320,328 retained responses. A separate execution supplied 3,672 replacement responses for 612 questions. Only the complete word eate changed to ate. All source labels stayed unchanged.

Correction: eate-to-ate-v1. Replacement execution: results/corrections/2026-11-06_15-37-32-64caf3fd-eate-to-ate-v1.

The 54,000 cases still have six responses each. Replaced responses are excluded from accuracy, coverage, format, and pairwise results. They remain in the historical cost ledger. The six new preflight calls are separate from corpus results.

Model	Cases	Changed answers	Matches before	Matches after	Strict before	Strict after
Claude Haiku 5.5	612	70	523	549	612	612
Claude Sonnet 5.5	612	2	611	611	533	543
Gemini 3.1 Flash-Lite Flex	612	4	610	608	612	612
Gemini 3.5 Flash-Lite Flex	612	30	575	583	611	612
GPT-4o mini	612	24	448	464	575	564
GPT-6 Luna Flex	612	42	545	547	612	612

Changes can reflect both corrected text and model or service variability. This comparison does not isolate a causal effect of the spelling correction.

Replacement audit · Costs by execution and replacement role · Before/after measures

Cost scope	Model	Generation IDs	Unknown-cost IDs	Known USD
retained_original_corpus	Claude Haiku 5.5	53388	352	0.782402
retained_original_corpus	Claude Sonnet 5.5	53388	0	26.415554
retained_original_corpus	Gemini 3.1 Flash-Lite Flex	53388	0	0.610217
retained_original_corpus	Gemini 3.5 Flash-Lite Flex	53388	0	0.750926
retained_original_corpus	GPT-4o mini	53388	0	0.796819
retained_original_corpus	GPT-6 Luna Flex	53813	425	0.318258
superseded_original_corpus	Claude Haiku 5.5	612	0	0.007434
superseded_original_corpus	Claude Sonnet 5.5	612	0	0.231660
superseded_original_corpus	Gemini 3.1 Flash-Lite Flex	612	0	0.005712
superseded_original_corpus	Gemini 3.5 Flash-Lite Flex	612	0	0.007070
superseded_original_corpus	GPT-4o mini	612	0	0.007550
superseded_original_corpus	GPT-6 Luna Flex	613	1	0.003121
original_preflight	Claude Haiku 5.5	1	0	0.000012
original_preflight	Claude Sonnet 5.5	1	0	0.000234
original_preflight	Gemini 3.1 Flash-Lite Flex	1	0	0.000009
original_preflight	Gemini 3.5 Flash-Lite Flex	1	0	0.000011
original_preflight	GPT-4o mini	1	0	0.000012
original_preflight	GPT-6 Luna Flex	1	0	0.000005
replacement_corpus	Claude Haiku 5.5	612	0	0.007424
replacement_corpus	Claude Sonnet 5.5	612	0	0.230440
replacement_corpus	Gemini 3.1 Flash-Lite Flex	612	0	0.005559
replacement_corpus	Gemini 3.5 Flash-Lite Flex	612	0	0.006885

replacement_corpus	GPT-4o mini	612	0	0.007373
replacement_corpus	GPT-6 Luna Flex	613	1	0.003060
correction_preflight	Claude Haiku 5.5	1	0	0.000011
correction_preflight	Claude Sonnet 5.5	1	0	0.000218
correction_preflight	Gemini 3.1 Flash-Lite Flex	1	0	0.000008
correction_preflight	Gemini 3.5 Flash-Lite Flex	1	0	0.000010
correction_preflight	GPT-4o mini	1	0	0.000011
correction_preflight	GPT-6 Luna Flex	1	0	0.000005

The replacement corpus phase took 3.19 minutes including closure. The operational time table below describes only the original execution. There is no single elapsed duration for the composite corpus.

Main findings

Six models answered the same 54,000 distinct questions. The principal results cover 108 model/dataset combinations; the correction used 12 replacement tasks. The corpus has 324,000 saved responses and zero final sample execution errors.
There were 288,516 source-label matches, 352 content filters, and 6 other non-interpretable responses across all models. These are separate outcomes.
Claude Sonnet 5.5 had the highest observed agreement with source labels: 53,973/53,995 (99.96%). This comparison concerns this corpus and this run. Source labels have not been independently audited for logical validity.

Sonnet 5.5 and Gemini 3.1 Flash-Lite are close in source-label agreement, but differ in format compliance and reported cost. See the tables below for exact counts and missing cost evidence.

Table of contents

Corpus correction and response replacement

Main findings

What question does the experiment ask?

What cases are in the corpus?

Which models answered?

How are answers counted?

What did each model return?

False positives and false negatives

Yes and no answers

Where do results differ?

Case family

Lexical condition

Verbal time

Premise order

Source validity label

Question formulation

Datasets and complete strata

Do models disagree on the same cases?

Cases that make the results concrete

HS: different answers

HS: content filter

HS: interpretable, non-strict format

mHS: different answers

mHS: content filter

mHS: interpretable, non-strict format

What did the run cost?

Tokens in final responses

What conclusions are supported?

Operational appendix: original execution time and concurrency

Request and returned metadata checks

Evidence, definitions, and reproduction

What question does the experiment ask?

We test how six language models answer hypothetical syllogism and metainferential hypothetical syllogism questions under different lexical and grammatical conditions. We compare each answer with the label supplied by the dataset. We also test whether the model follows a strict response-format instruction.

Every case uses a new conversation, one system message, and one original question. No examples are supplied in the prompt (zero-shot). The system message is:

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Answer only with 'yes' or 'no' and nothing else.
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A case is one complete reasoning question. It can contain several clauses or sentences. We count cases, not individual clauses. The 54,000 question texts are distinct by exact text comparison.

What cases are in the corpus?

Code	Meaning	Cases
HS	Hypothetical syllogism expressed with conditionals	18,000
mHS	Metainferential hypothetical syllogism: relations between inferences	36,000
HF / LF / NS	High-frequency words / low-frequency words / pseudowords	18,000 each
PR / PA / FU	Present / past / future	18,000 each
canonic / non-canonic	Standard / reversed premise order, as labeled by the source	27,000 each
valid / invalid	Source validity label	27,000 each

NS is a pseudoword condition, not another level of word frequency. The filename identifies HS versus mHS; the original internal case-type field is HS in both families. We preserve that field and report family separately.

Each HS dataset has four strata: source validity crossed with premise order. Each mHS dataset has eight strata: those four combinations crossed with two formulations. The formulations are assuming that we can infer / thus and from the inference (therefore) / does the inference follow. Each stratum contains 500 cases. The complete design has 108 strata.

Dataset	Family	Lexical condition	Tense	Cases	Strata
HS_HF_FU_2000	HS	HF	FU	2000	4
HS_HF_PA_2000	HS	HF	PA	2000	4
HS_HF_PR_2000	HS	HF	PR	2000	4
HS_LF_FU_2000	HS	LF	FU	2000	4
HS_LF_PA_2000	HS	LF	PA	2000	4
HS_LF_PR_2000	HS	LF	PR	2000	4
HS_NS_FU_2000	HS	NS	FU	2000	4
HS_NS_PA_2000	HS	NS	PA	2000	4
HS_NS_PR_2000	HS	NS	PR	2000	4
mHS_HF_FU_4000	mHS	HF	FU	4000	8
mHS_HF_PA_4000	mHS	HF	PA	4000	8
mHS_HF_PR_4000	mHS	HF	PR	4000	8
mHS_LF_FU_4000	mHS	LF	FU	4000	8
mHS_LF_PA_4000	mHS	LF	PA	4000	8
mHS_LF_PR_4000	mHS	LF	PR	4000	8
mHS_NS_FU_4000	mHS	NS	FU	4000	8
mHS_NS_PA_4000	mHS	NS	PA	4000	8

mHS_NS_PR_4000	mHS	NS	PR	4000	8

mHS accounts for two thirds of the corpus. Overall results weight each case equally, so mHS has twice the weight of HS. Family-specific results are shown separately. The source labels are balanced: 27,000 valid and 27,000 invalid cases.

Working inputs preserve the original question text, order, source identifiers, and all record fields, except the approved translations of valido to valid and canonico to canonic in the two LF/PR datasets. The analysis verifies every saved prompt and target against the frozen inputs and the Git source revision.

Which models answered?

The exact model IDs identify the experiment. Provider, service tier, and reasoning settings are part of each model condition. These differences prevent us from attributing differences to a model name or reasoning setting alone.

Model ID	Fixed endpoint	Tier	Reasoning requested	Effort
openai/gpt-4o-mini	openai	default	not supported	omitted
openai/gpt-6-luna	openai/flex	flex	disabled	omitted
anthropic/claude-haiku-5.5	anthropic	default	disabled	medium
anthropic/claude-sonnet-5.5	anthropic	default	enabled	low
google/gemini-3.1-flash-lite	google-ai-studio/flex	flex	disabled	omitted
google/gemini-3.5-flash-lite	google-ai-studio/flex	flex	enabled	minimal

GPT-4o mini: archived model card

GPT-6 Luna Flex: archived model card

Claude Haiku 5.5: archived model card

Claude Sonnet 5.5: archived model card

Gemini 3.1 Flash-Lite Flex: archived model card

Gemini 3.5 Flash-Lite Flex: archived model card

Providers were fixed and fallbacks disabled. The output limit was 1,024 tokens, including reasoning where the provider counts it in the output budget. Temperature, top-p, and seed were omitted. Responses are not deterministic. Caching and streaming were disabled.

The archived earlier and later tags are comparison slots. They are not equivalent generations across developers. Haiku and Sonnet are different model families. The main results therefore compare six individual models.

How are answers counted?

Measure	Numerator	Denominator
Source-label agreement	Matches	Interpretable answers
Interpretation coverage	Interpretable answers	All planned cases
Strict-format compliance	Only yes or no, ignoring case and outer whitespace	All planned cases
Matches over all cases	Matches	All planned cases

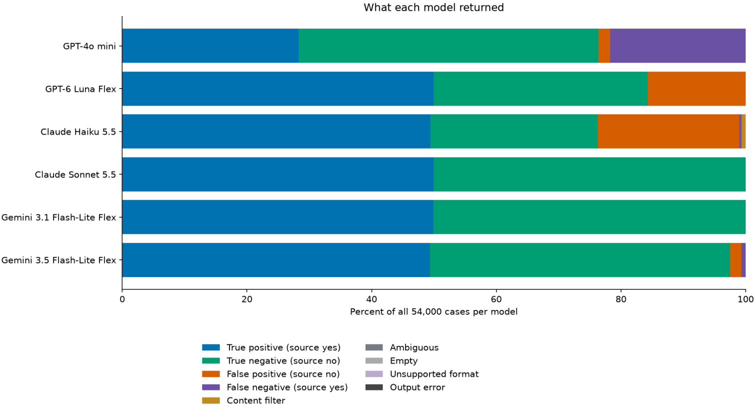
Scoring uses source-label-deterministic-v3 and extraction rule initial_or_final_line_v2. It can read a supported yes/no answer at the beginning or on the final line. Explanations, punctuation, and bold answers may be interpretable but still fail the strict-format instruction. A response containing both complete terms yes and no is ambiguous. Errors, filters, empty responses, ambiguity, and unsupported formats keep their existing precedence. They are never converted to no.

A source-label disagreement is not automatically a logical error. A source-label match is not independent proof of logical validity. We do not change source labels or scores in this report.

What did each model return?

Model	Agreement / interpreted	Coverage / 54,000	Strict / 54,000	Matches / 54,000
GPT-4o mini	41,234/53,999 (76.36%)	53,999/54,000 (100.00%)	50,235/54,000 (93.03%)	41,234/54,000 (76.36%)
GPT-6 Luna Flex	45,492/54,000 (84.24%)	54,000/54,000 (100.00%)	54,000/54,000 (100.00%)	45,492/54,000 (84.24%)
Claude Haiku 5.5	41,212/53,648 (76.82%)	53,648/54,000 (99.35%)	53,642/54,000 (99.34%)	41,212/54,000 (76.32%)
Claude Sonnet 5.5	53,973/53,995 (99.96%)	53,995/54,000 (99.99%)	44,350/54,000 (82.13%)	53,973/54,000 (99.95%)
Gemini 3.1 Flash-Lite Flex	53,957/54,000 (99.92%)	54,000/54,000 (100.00%)	53,961/54,000 (99.93%)	53,957/54,000 (99.92%)
Gemini 3.5 Flash-Lite Flex	52,648/54,000 (97.50%)	54,000/54,000 (100.00%)	53,988/54,000 (99.98%)	52,648/54,000 (97.50%)

Model	Matches	Disagreements	FP	FN	Filters	Empty	Ambiguous	Unsupported format	Errors
GPT-4o mini	41,234	12,765	1,035	11,730	0	0	0	1	0
GPT-6 Luna Flex	45,492	8,508	8,490	18	0	0	0	0	0
Claude Haiku 5.5	41,212	12,436	12,237	199	352	0	0	0	0
Claude Sonnet 5.5	53,973	22	0	22	0	0	5	0	0
Gemini 3.1 Flash-Lite Flex	53,957	43	0	43	0	0	0	0	0
Gemini 3.5 Flash-Lite Flex	52,648	1,352	987	365	0	0	0	0	0



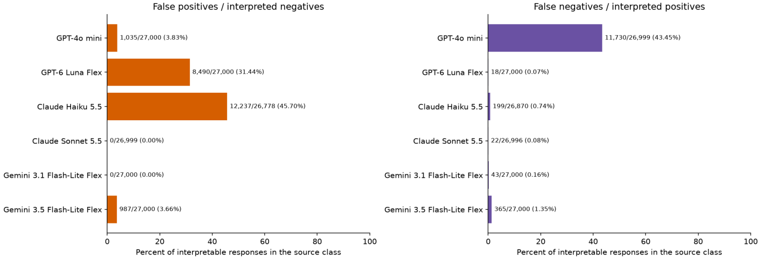
False positives and false negatives

Positive means yes. FP is an interpreted yes when the source target is no; FN is an interpreted no when the source target is yes. TP and TN match the source target. These terms refer to source labels, not audited logical truth. Filters and other non-interpretable outcomes remain separate.

FPR = FP / (FP + TN), among interpreted source-negative cases. FNR = FN / (FN + TP), among interpreted source-positive cases. Class coverage uses every planned case of that source class. Undefined means that the denominator is zero; it does not mean a zero error rate.

Model	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	26671	14541	12237	199	12,237/26,778 (45.70%)	199/26,870 (0.74%)	26,778/27,000 (99.18%)	26,870/27,000 (99.52%)
Claude Sonnet 5.5	26974	26999	0	22	0/26,999 (0.00%)	22/26,996 (0.08%)	26,999/27,000 (100.00%)	26,996/27,000 (99.99%)
Gemini 3.1 Flash-Lite Flex	26957	27000	0	43	0/27,000 (0.00%)	43/27,000 (0.16%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	26635	26013	987	365	987/27,000 (3.66%)	365/27,000 (1.35%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-4o mini	15269	25965	1035	11730	1,035/27,000 (3.83%)	11,730/26,999 (43.45%)	27,000/27,000 (100.00%)	26,999/27,000 (100.00%)
GPT-6 Luna Flex	26982	18510	8490	18	8,490/27,000 (31.44%)	18/27,000 (0.07%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)

Direction of disagreement: yes is positive; source labels are not audited truth



Haiku and Luna have more false positives than false negatives in this run. GPT-4o mini has more false negatives. Sonnet and Gemini 3.1 have no false positives among their interpreted responses; their disagreements are false negatives. This describes the direction of source-label disagreement, not its cause.

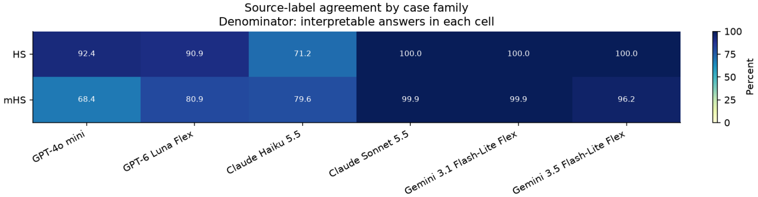
Yes and no answers

Model	Interpreted answer	Count
Claude Haiku 5.5	no	14740
Claude Haiku 5.5	yes	38908
Claude Sonnet 5.5	no	27021
Claude Sonnet 5.5	yes	26974
Gemini 3.1 Flash-Lite Flex	no	27043
Gemini 3.1 Flash-Lite Flex	yes	26957
Gemini 3.5 Flash-Lite Flex	no	26378
Gemini 3.5 Flash-Lite Flex	yes	27622
GPT-4o mini	no	37695
GPT-4o mini	yes	16304
GPT-6 Luna Flex	no	18528
GPT-6 Luna Flex	yes	35472

The source-label confusion table crosses the expected answer with the interpreted answer. It excludes non-interpretable responses, whose counts remain above.

Where do results differ?

The following tables compare observed results within the corpus. They do not establish causal effects. Each cell states its denominator. Comparisons between models use the same source cases; comparisons between conditions do not assume that cases are matched pairs.



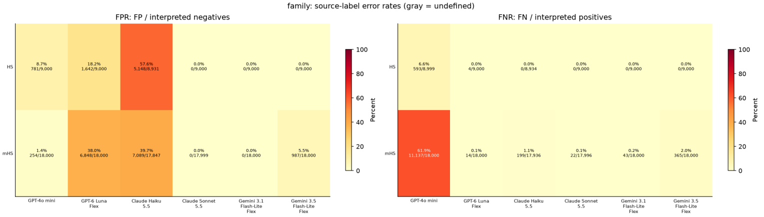
Case family

The largest observed range across case family is for GPT-4o mini: 24.01 percentage points between HS (16,625/17,999 (92.37%)) and mHS (24,609/36,000 (68.36%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format

Claude Haiku 5.5	HS	12,717/17,865 (71.18%)	17,865/18,000 (99.25%)	17,864/18,000 (99.24%)
Claude Haiku 5.5	mHS	28,495/35,783 (79.63%)	35,783/36,000 (99.40%)	35,778/36,000 (99.38%)
Claude Sonnet 5.5	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	12,529/18,000 (69.61%)
Claude Sonnet 5.5	mHS	35,973/35,995 (99.94%)	35,995/36,000 (99.99%)	31,821/36,000 (88.39%)
Gemini 3.1 Flash-Lite Flex	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.1 Flash-Lite Flex	mHS	35,957/36,000 (99.88%)	36,000/36,000 (100.00%)	35,964/36,000 (99.90%)
Gemini 3.5 Flash-Lite Flex	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS	34,648/36,000 (96.24%)	36,000/36,000 (100.00%)	35,988/36,000 (99.97%)
GPT-4o mini	HS	16,625/17,999 (92.37%)	17,999/18,000 (99.99%)	16,480/18,000 (91.56%)
GPT-4o mini	mHS	24,609/36,000 (68.36%)	36,000/36,000 (100.00%)	33,755/36,000 (93.76%)
GPT-6 Luna Flex	HS	16,354/18,000 (90.86%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	mHS	29,138/36,000 (80.94%)	36,000/36,000 (100.00%)	36,000/36,000 (100.00%)

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	HS	8934	3783	5148	0	5,148/8,931 (57.64%)	0/8,934 (0.00%)	8,931/9,000 (99.23%)	8,934/9,000 (99.27%)
Claude Haiku 5.5	mHS	17737	10758	7089	199	7,089/17,847 (39.72%)	199/17,936 (1.11%)	17,847/18,000 (99.15%)	17,936/18,000 (99.64%)
Claude Sonnet 5.5	HS	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Claude Sonnet 5.5	mHS	17974	17999	0	22	0/17,999 (0.00%)	22/17,996 (0.12%)	17,999/18,000 (99.99%)	17,996/18,000 (99.98%)
Gemini 3.1 Flash-Lite Flex	HS	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS	17957	18000	0	43	0/18,000 (0.00%)	43/18,000 (0.24%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS	17635	17013	987	365	987/18,000 (5.48%)	365/18,000 (2.03%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-4o mini	HS	8406	8219	781	593	781/9,000 (8.68%)	593/8,999 (6.59%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
GPT-4o mini	mHS	6863	17746	254	11137	254/18,000 (1.41%)	11,137/18,000 (61.87%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	HS	8996	7358	1642	4	1,642/9,000 (18.24%)	4/9,000 (0.04%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	mHS	17986	11152	6848	14	6,848/18,000 (38.04%)	14/18,000 (0.08%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)



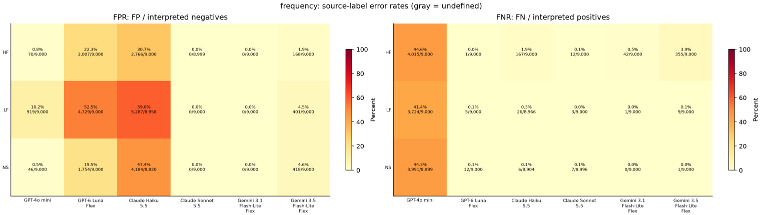
Lexical condition

The largest observed range across lexical condition is for GPT-6 Luna Flex: 16.49 percentage points between NS (16,234/18,000 (90.19%)) and LF (13,266/18,000 (73.70%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	HF	15,067/18,000 (83.71%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Claude Haiku 5.5	LF	12,611/17,924 (70.36%)	17,924/18,000 (99.58%)	17,920/18,000 (99.56%)
Claude Haiku 5.5	NS	13,534/17,724 (76.36%)	17,724/18,000 (98.47%)	17,723/18,000 (98.46%)
Claude Sonnet 5.5	HF	17,987/17,999 (99.93%)	17,999/18,000 (99.99%)	15,540/18,000 (86.33%)

Claude Sonnet 5.5	LF	17,997/18,000 (99.98%)	18,000/18,000 (100.00%)	14,165/18,000 (78.69%)
Claude Sonnet 5.5	NS	17,989/17,996 (99.96%)	17,996/18,000 (99.98%)	14,645/18,000 (81.36%)
Gemini 3.1 Flash-Lite Flex	HF	17,958/18,000 (99.77%)	18,000/18,000 (100.00%)	17,970/18,000 (99.83%)
Gemini 3.1 Flash-Lite Flex	LF	17,999/18,000 (99.99%)	18,000/18,000 (100.00%)	17,992/18,000 (99.96%)
Gemini 3.1 Flash-Lite Flex	NS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	HF	17,477/18,000 (97.09%)	18,000/18,000 (100.00%)	17,990/18,000 (99.94%)
Gemini 3.5 Flash-Lite Flex	LF	17,590/18,000 (97.72%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	NS	17,581/18,000 (97.67%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
GPT-4o mini	HF	13,915/18,000 (77.31%)	18,000/18,000 (100.00%)	16,274/18,000 (90.41%)
GPT-4o mini	LF	13,357/18,000 (74.21%)	18,000/18,000 (100.00%)	16,725/18,000 (92.92%)
GPT-4o mini	NS	13,962/17,999 (77.57%)	17,999/18,000 (99.99%)	17,236/18,000 (95.76%)
GPT-6 Luna Flex	HF	15,992/18,000 (88.84%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	LF	13,266/18,000 (73.70%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	NS	16,234/18,000 (90.19%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	HF	8833	6234	2766	167	2,766/9,000 (30.73%)	167/9,000 (1.86%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Claude Haiku 5.5	LF	8940	3671	5287	26	5,287/8,958 (59.02%)	26/8,966 (0.29%)	8,958/9,000 (99.53%)	8,966/9,000 (99.62%)
Claude Haiku 5.5	NS	8898	4636	4184	6	4,184/8,820 (47.44%)	6/8,904 (0.07%)	8,820/9,000 (98.00%)	8,904/9,000 (98.93%)
Claude Sonnet 5.5	HF	8988	8999	0	12	0/8,999 (0.00%)	12/9,000 (0.13%)	8,999/9,000 (99.99%)	9,000/9,000 (100.00%)
Claude Sonnet 5.5	LF	8997	9000	0	3	0/9,000 (0.00%)	3/9,000 (0.03%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Claude Sonnet 5.5	NS	8989	9000	0	7	0/9,000 (0.00%)	7/8,996 (0.08%)	9,000/9,000 (100.00%)	8,996/9,000 (99.96%)
Gemini 3.1 Flash-Lite Flex	HF	8958	9000	0	42	0/9,000 (0.00%)	42/9,000 (0.47%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	LF	8999	9000	0	1	0/9,000 (0.00%)	1/9,000 (0.01%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	NS	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HF	8645	8832	168	355	168/9,000 (1.87%)	355/9,000 (3.94%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	LF	8991	8599	401	9	401/9,000 (4.46%)	9/9,000 (0.10%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	NS	8999	8582	418	1	418/9,000 (4.64%)	1/9,000 (0.01%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	HF	4985	8930	70	4015	70/9,000 (0.78%)	4,015/9,000 (44.61%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	LF	5276	8081	919	3724	919/9,000 (10.21%)	3,724/9,000 (41.38%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	NS	5008	8954	46	3991	46/9,000 (0.51%)	3,991/8,999 (44.35%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
GPT-6 Luna Flex	HF	8999	6993	2007	1	2,007/9,000 (22.30%)	1/9,000 (0.01%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	LF	8995	4271	4729	5	4,729/9,000 (52.54%)	5/9,000 (0.06%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	NS	8988	7246	1754	12	1,754/9,000 (19.49%)	12/9,000 (0.13%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)



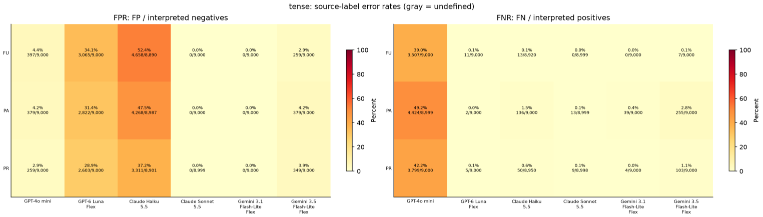
Verbal time

The largest observed range across verbal time is for Claude Haiku 5.5: 7.40 percentage points between PR (14,490/17,851 (81.17%)) and FU (13,139/17,810 (73.77%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	FU	13,139/17,810 (73.77%)	17,810/18,000 (98.94%)	17,805/18,000 (98.92%)
Claude Haiku 5.5	PA	13,583/17,987 (75.52%)	17,987/18,000 (99.93%)	17,986/18,000 (99.92%)
Claude Haiku 5.5	PR	14,490/17,851 (81.17%)	17,851/18,000 (99.17%)	17,851/18,000 (99.17%)
Claude Sonnet 5.5	FU	17,999/17,999 (100.00%)	17,999/18,000 (99.99%)	14,529/18,000 (80.72%)
Claude Sonnet 5.5	PA	17,986/17,999 (99.93%)	17,999/18,000 (99.99%)	14,555/18,000 (80.86%)
Claude Sonnet 5.5	PR	17,988/17,997 (99.95%)	17,997/18,000 (99.98%)	15,266/18,000 (84.81%)
Gemini 3.1 Flash-Lite Flex	FU	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,987/18,000 (99.93%)
Gemini 3.1 Flash-Lite Flex	PA	17,961/18,000 (99.78%)	18,000/18,000 (100.00%)	17,990/18,000 (99.94%)
Gemini 3.1 Flash-Lite Flex	PR	17,996/18,000 (99.98%)	18,000/18,000 (100.00%)	17,984/18,000 (99.91%)
Gemini 3.5 Flash-Lite Flex	FU	17,734/18,000 (98.52%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	PA	17,366/18,000 (96.48%)	18,000/18,000 (100.00%)	17,994/18,000 (99.97%)
Gemini 3.5 Flash-Lite Flex	PR	17,548/18,000 (97.49%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
GPT-4o mini	FU	14,096/18,000 (78.31%)	18,000/18,000 (100.00%)	16,660/18,000 (92.56%)
GPT-4o mini	PA	13,196/17,999 (73.32%)	17,999/18,000 (99.99%)	17,021/18,000 (94.56%)
GPT-4o mini	PR	13,942/18,000 (77.46%)	18,000/18,000 (100.00%)	16,554/18,000 (91.97%)
GPT-6 Luna Flex	FU	14,924/18,000 (82.91%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	PA	15,176/18,000 (84.31%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	PR	15,392/18,000 (85.51%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	FU	8907	4232	4658	13	4,658/8,890 (52.40%)	13/8,920 (0.15%)	8,890/9,000 (98.78%)	8,920/9,000 (99.11%)
Claude Haiku 5.5	PA	8864	4719	4268	136	4,268/8,987 (47.49%)	136/9,000 (1.51%)	8,987/9,000 (99.86%)	9,000/9,000 (100.00%)
Claude Haiku 5.5	PR	8900	5590	3311	50	3,311/8,901 (37.20%)	50/8,950 (0.56%)	8,901/9,000 (98.90%)	8,950/9,000 (99.44%)
Claude Sonnet 5.5	FU	8999	9000	0	0	0/9,000 (0.00%)	0/8,999 (0.00%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
Claude Sonnet 5.5	PA	8986	9000	0	13	0/9,000 (0.00%)	13/8,999 (0.14%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
Claude Sonnet 5.5	PR	8989	8999	0	9	0/8,999 (0.00%)	9/8,998 (0.10%)	8,999/9,000 (99.99%)	8,998/9,000 (99.98%)
Gemini 3.1 Flash-Lite Flex	FU	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	PA	8961	9000	0	39	0/9,000 (0.00%)	39/9,000 (0.43%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	PR	8996	9000	0	4	0/9,000 (0.00%)	4/9,000 (0.04%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	FU	8993	8741	259	7	259/9,000 (2.88%)	7/9,000 (0.08%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	PA	8745	8621	379	255	379/9,000 (4.21%)	255/9,000 (2.83%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)

Gemini 3.5 Flash-Lite Flex	PR	8897	8651	349	103	349/9,000 (3.88%)	103/9,000 (1.14%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	FU	5493	8603	397	3507	397/9,000 (4.41%)	3,507/9,000 (38.97%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	PA	4575	8621	379	4424	379/9,000 (4.21%)	4,424/8,999 (49.16%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
GPT-4o mini	PR	5201	8741	259	3799	259/9,000 (2.88%)	3,799/9,000 (42.21%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	FU	8989	5935	3065	11	3,065/9,000 (34.06%)	11/9,000 (0.12%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	PA	8998	6178	2822	2	2,822/9,000 (31.36%)	2/9,000 (0.02%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	PR	8995	6397	2603	5	2,603/9,000 (28.92%)	5/9,000 (0.06%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)



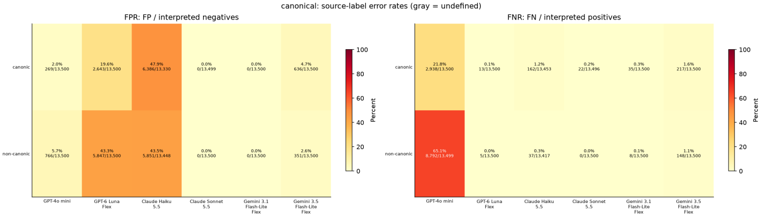
Premise order

The largest observed range across premise order is for GPT-4o mini: 23.52 percentage points between canonic (23,793/27,000 (88.12%)) and non-canonic (17,441/26,999 (64.60%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	canonic	20,235/26,783 (75.55%)	26,783/27,000 (99.20%)	26,783/27,000 (99.20%)
Claude Haiku 5.5	non-canonic	20,977/26,865 (78.08%)	26,865/27,000 (99.50%)	26,859/27,000 (99.48%)
Claude Sonnet 5.5	canonic	26,973/26,995 (99.92%)	26,995/27,000 (99.98%)	23,587/27,000 (87.36%)
Claude Sonnet 5.5	non-canonic	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)	20,763/27,000 (76.90%)
Gemini 3.1 Flash-Lite Flex	canonic	26,965/27,000 (99.87%)	27,000/27,000 (100.00%)	26,968/27,000 (99.88%)
Gemini 3.1 Flash-Lite Flex	non-canonic	26,992/27,000 (99.97%)	27,000/27,000 (100.00%)	26,993/27,000 (99.97%)
Gemini 3.5 Flash-Lite Flex	canonic	26,147/27,000 (96.84%)	27,000/27,000 (100.00%)	26,988/27,000 (99.96%)
Gemini 3.5 Flash-Lite Flex	non-canonic	26,501/27,000 (98.15%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-4o mini	canonic	23,793/27,000 (88.12%)	27,000/27,000 (100.00%)	24,920/27,000 (92.30%)
GPT-4o mini	non-canonic	17,441/26,999 (64.60%)	26,999/27,000 (100.00%)	25,315/27,000 (93.76%)
GPT-6 Luna Flex	canonic	24,344/27,000 (90.16%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-6 Luna Flex	non-canonic	21,148/27,000 (78.33%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	canonic	13291	6944	6386	162	6,386/13,330 (47.91%)	162/13,453 (1.20%)	13,330/13,500 (98.74%)	13,453/13,500 (99.65%)
Claude Haiku 5.5	non-canonic	13380	7597	5851	37	5,851/13,448 (43.51%)	37/13,417 (0.28%)	13,448/13,500 (99.61%)	13,417/13,500 (99.39%)
Claude Sonnet 5.5	canonic	13474	13499	0	22	0/13,499 (0.00%)	22/13,496 (0.16%)	13,499/13,500 (99.99%)	13,496/13,500 (99.97%)
Claude Sonnet 5.5	non-canonic	13500	13500	0	0	0/13,500 (0.00%)	0/13,500 (0.00%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
Gemini 3.1 Flash-Lite Flex	canonic	13465	13500	0	35	0/13,500 (0.00%)	35/13,500 (0.26%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
Gemini 3.1 Flash-Lite Flex	non-canonic	13492	13500	0	8	0/13,500 (0.00%)	8/13,500 (0.06%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
Gemini 3.5 Flash-Lite Flex	canonic	13283	12864	636	217	636/13,500 (4.71%)	217/13,500 (1.61%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)

Gemini 3.5 Flash-Lite Flex	non-canonic	13352	13149	351	148	351/13,500 (2.60%)	148/13,500 (1.10%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
GPT-4o mini	canonic	10562	13231	269	2938	269/13,500 (1.99%)	2,938/13,500 (21.76%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
GPT-4o mini	non-canonic	4707	12734	766	8792	766/13,500 (5.67%)	8,792/13,499 (65.13%)	13,500/13,500 (100.00%)	13,499/13,500 (99.99%)
GPT-6 Luna Flex	canonic	13487	10857	2643	13	2,643/13,500 (19.58%)	13/13,500 (0.10%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)
GPT-6 Luna Flex	non-canonic	13495	7653	5847	5	5,847/13,500 (43.31%)	5/13,500 (0.04%)	13,500/13,500 (100.00%)	13,500/13,500 (100.00%)



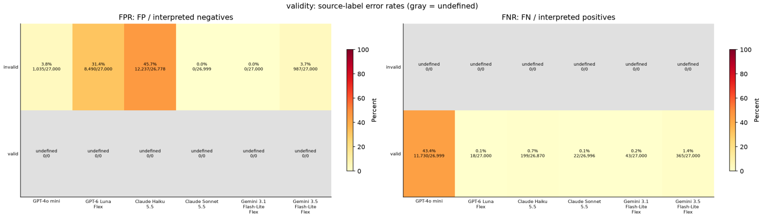
Source validity label

The largest observed range across source validity label is for Claude Haiku 5.5: 44.96 percentage points between valid (26,671/26,870 (99.26%)) and invalid (14,541/26,778 (54.30%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	invalid	14,541/26,778 (54.30%)	26,778/27,000 (99.18%)	26,773/27,000 (99.16%)
Claude Haiku 5.5	valid	26,671/26,870 (99.26%)	26,870/27,000 (99.52%)	26,869/27,000 (99.51%)
Claude Sonnet 5.5	invalid	26,999/26,999 (100.00%)	26,999/27,000 (100.00%)	17,612/27,000 (65.23%)
Claude Sonnet 5.5	valid	26,974/26,996 (99.92%)	26,996/27,000 (99.99%)	26,738/27,000 (99.03%)
Gemini 3.1 Flash-Lite Flex	invalid	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)	26,967/27,000 (99.88%)
Gemini 3.1 Flash-Lite Flex	valid	26,957/27,000 (99.84%)	27,000/27,000 (100.00%)	26,994/27,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	invalid	26,013/27,000 (96.34%)	27,000/27,000 (100.00%)	26,994/27,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	valid	26,635/27,000 (98.65%)	27,000/27,000 (100.00%)	26,994/27,000 (99.98%)
GPT-4o mini	invalid	25,965/27,000 (96.17%)	27,000/27,000 (100.00%)	26,125/27,000 (96.76%)
GPT-4o mini	valid	15,269/26,999 (56.55%)	26,999/27,000 (100.00%)	24,110/27,000 (89.30%)
GPT-6 Luna Flex	invalid	18,510/27,000 (68.56%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-6 Luna Flex	valid	26,982/27,000 (99.93%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	invalid	0	14541	12237	0	12,237/26,778 (45.70%)	0/0 (undefined)	26,778/27,000 (99.18%)	0/0 (undefined)
Claude Haiku 5.5	valid	26671	0	0	199	0/0 (undefined)	199/26,870 (0.74%)	0/0 (undefined)	26,870/27,000 (99.52%)
Claude Sonnet 5.5	invalid	0	26999	0	0	0/26,999 (0.00%)	0/0 (undefined)	26,999/27,000 (100.00%)	0/0 (undefined)
Claude Sonnet 5.5	valid	26974	0	0	22	0/0 (undefined)	22/26,996 (0.08%)	0/0 (undefined)	26,996/27,000 (99.99%)
Gemini 3.1 Flash-Lite Flex	invalid	0	27000	0	0	0/27,000 (0.00%)	0/0 (undefined)	27,000/27,000 (100.00%)	0/0 (undefined)
Gemini 3.1 Flash-Lite Flex	valid	26957	0	0	43	0/0 (undefined)	43/27,000 (0.16%)	0/0 (undefined)	27,000/27,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	invalid	0	26013	987	0	987/27,000 (3.66%)	0/0 (undefined)	27,000/27,000 (100.00%)	0/0 (undefined)
Gemini 3.5 Flash-Lite Flex	valid	26635	0	0	365	0/0 (undefined)	365/27,000 (1.35%)	0/0 (undefined)	27,000/27,000 (100.00%)
GPT-4o mini	invalid	0	25965	1035	0	1,035/27,000 (3.83%)	0/0 (undefined)	27,000/27,000 (100.00%)	0/0 (undefined)
GPT-4o mini	valid	15269	0	0	11730	0/0 (undefined)	11,730/26,999 (43.45%)	0/0 (undefined)	26,999/27,000 (100.00%)

GPT-6 Luna Flex	invalid	0	18510	8490	0	8,490/27,000 (31.44%)	0/0 (undefined)	27,000/27,000 (100.00%)	0/0 (undefined)
GPT-6 Luna Flex	valid	26982	0	0	18	0/0 (undefined)	18/27,000 (0.07%)	0/0 (undefined)	27,000/27,000 (100.00%)



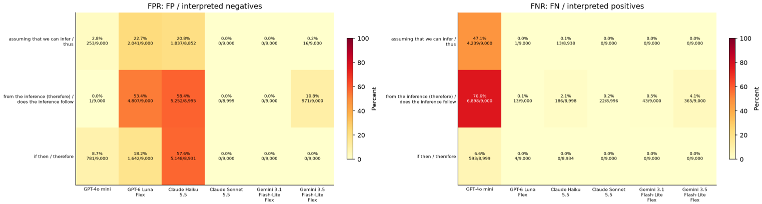
Question formulation

The largest observed range across question formulation is for GPT-4o mini: 30.69 percentage points between if then / therefore (16,625/17,999 (92.37%)) and from the inference (therefore) / does the inference follow (11,101/18,000 (61.67%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	assuming that we can infer / thus	15,940/17,790 (89.60%)	17,790/18,000 (98.83%)	17,789/18,000 (98.83%)
Claude Haiku 5.5	from the inference (therefore) / does the inference follow	12,555/17,993 (69.78%)	17,993/18,000 (99.96%)	17,989/18,000 (99.94%)
Claude Haiku 5.5	if then / therefore	12,717/17,865 (71.18%)	17,865/18,000 (99.25%)	17,864/18,000 (99.24%)
Claude Sonnet 5.5	assuming that we can infer / thus	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	15,849/18,000 (88.05%)
Claude Sonnet 5.5	from the inference (therefore) / does the inference follow	17,973/17,995 (99.88%)	17,995/18,000 (99.97%)	15,972/18,000 (88.73%)
Claude Sonnet 5.5	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	12,529/18,000 (69.61%)
Gemini 3.1 Flash-Lite Flex	assuming that we can infer / thus	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,975/18,000 (99.86%)
Gemini 3.1 Flash-Lite Flex	from the inference (therefore) / does the inference follow	17,957/18,000 (99.76%)	18,000/18,000 (100.00%)	17,989/18,000 (99.94%)
Gemini 3.1 Flash-Lite Flex	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	assuming that we can infer / thus	17,984/18,000 (99.91%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	from the inference (therefore) / does the inference follow	16,664/18,000 (92.58%)	18,000/18,000 (100.00%)	17,989/18,000 (99.94%)
Gemini 3.5 Flash-Lite Flex	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-4o mini	assuming that we can infer / thus	13,508/18,000 (75.04%)	18,000/18,000 (100.00%)	16,668/18,000 (92.60%)
GPT-4o mini	from the inference (therefore) / does the inference follow	11,101/18,000 (61.67%)	18,000/18,000 (100.00%)	17,087/18,000 (94.93%)
GPT-4o mini	if then / therefore	16,625/17,999 (92.37%)	17,999/18,000 (99.99%)	16,480/18,000 (91.56%)
GPT-6 Luna Flex	assuming that we can infer / thus	15,958/18,000 (88.66%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	from the inference (therefore) / does the inference follow	13,180/18,000 (73.22%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	if then / therefore	16,354/18,000 (90.86%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

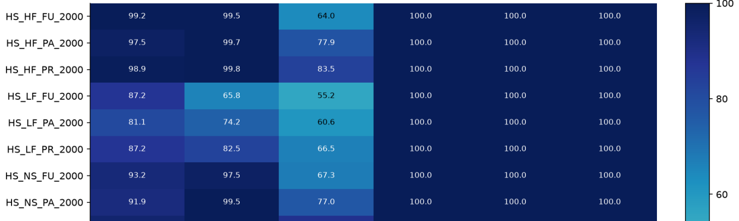
Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	assuming that we can infer / thus	8925	7015	1837	13	1,837/8,852 (20.75%)	13/8,938 (0.15%)	8,852/9,000 (98.36%)	8,938/9,000 (99.31%)
Claude Haiku 5.5	from the inference (therefore) / does the inference follow	8812	3743	5252	186	5,252/8,995 (58.39%)	186/8,998 (2.07%)	8,995/9,000 (99.94%)	8,998/9,000 (99.98%)
Claude Haiku 5.5	if then / therefore	8934	3783	5148	0	5,148/8,931 (57.64%)	0/8,934 (0.00%)	8,931/9,000 (99.23%)	8,934/9,000 (99.27%)
Claude Sonnet 5.5	assuming that we can infer / thus	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)

Claude Sonnet 5.5	from the inference (therefore) / does the inference follow	8974	8999	0	22	0/8,999 (0.00%)	22/8,996 (0.24%)	8,999/9,000 (99.99%)	8,996/9,000 (99.96%)
Claude Sonnet 5.5	if then / therefore	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	assuming that we can infer / thus	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	from the inference (therefore) / does the inference follow	8957	9000	0	43	0/9,000 (0.00%)	43/9,000 (0.48%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	if then / therefore	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	assuming that we can infer / thus	9000	8984	16	0	16/9,000 (0.18%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	from the inference (therefore) / does the inference follow	8635	8029	971	365	971/9,000 (10.79%)	365/9,000 (4.06%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	if then / therefore	9000	9000	0	0	0/9,000 (0.00%)	0/9,000 (0.00%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	assuming that we can infer / thus	4761	8747	253	4239	253/9,000 (2.81%)	4,239/9,000 (47.10%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	from the inference (therefore) / does the inference follow	2102	8999	1	6898	1/9,000 (0.01%)	6,898/9,000 (76.64%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-4o mini	if then / therefore	8406	8219	781	593	781/9,000 (8.68%)	593/8,999 (6.59%)	9,000/9,000 (100.00%)	8,999/9,000 (99.99%)
GPT-6 Luna Flex	assuming that we can infer / thus	8999	6959	2041	1	2,041/9,000 (22.68%)	1/9,000 (0.01%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	from the inference (therefore) / does the inference follow	8987	4193	4807	13	4,807/9,000 (53.41%)	13/9,000 (0.14%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)
GPT-6 Luna Flex	if then / therefore	8996	7358	1642	4	1,642/9,000 (18.24%)	4/9,000 (0.04%)	9,000/9,000 (100.00%)	9,000/9,000 (100.00%)

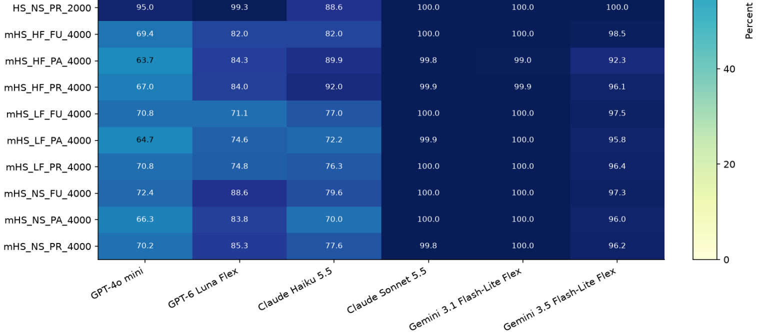


formulation: source-label error rates (gray = undefined)

Datasets and complete strata



Source-label agreement by dataset
Denominator: interpretable answers in each cell



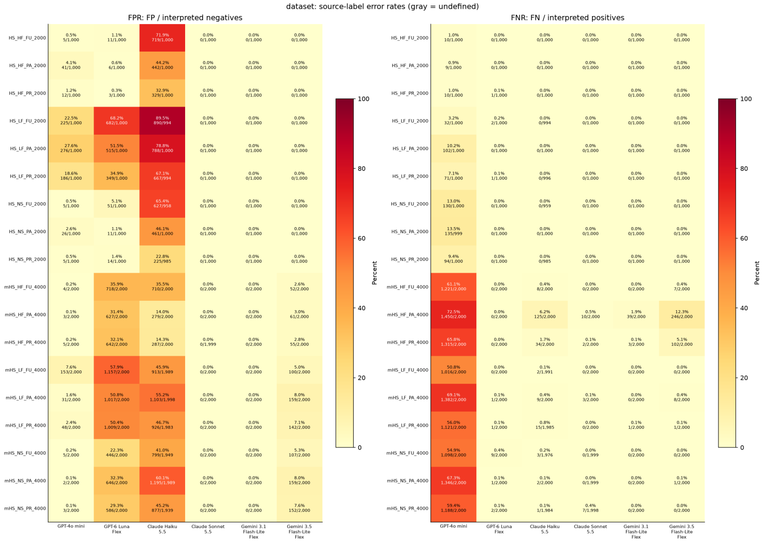
The dataset table contains all 108 model/dataset cells. The stratum results contain all 648 model/stratum cells. Join them to the stratum definitions by stratum. The HTML report supports combined condition filters.

Model	Condition	TP	TN	FP	FN	FPR: FP / negatives interpreted	FNR: FN / positives interpreted	Negative coverage	Positive coverage
Claude Haiku 5.5	HS_HF_FU_2000	1000	281	719	0	719/1,000 (71.90%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Haiku 5.5	HS_HF_PA_2000	1000	558	442	0	442/1,000 (44.20%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Haiku 5.5	HS_HF_PR_2000	1000	671	329	0	329/1,000 (32.90%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Haiku 5.5	HS_LF_FU_2000	994	104	890	0	890/994 (89.54%)	0/994 (0.00%)	994/1,000 (99.40%)	994/1,000 (99.40%)
Claude Haiku 5.5	HS_LF_PA_2000	1000	212	788	0	788/1,000 (78.80%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Haiku 5.5	HS_LF_PR_2000	996	327	667	0	667/994 (67.10%)	0/996 (0.00%)	994/1,000 (99.40%)	996/1,000 (99.60%)
Claude Haiku 5.5	HS_NS_FU_2000	959	331	627	0	627/958 (65.45%)	0/959 (0.00%)	958/1,000 (95.80%)	959/1,000 (95.90%)
Claude Haiku 5.5	HS_NS_PA_2000	1000	539	461	0	461/1,000 (46.10%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Haiku 5.5	HS_NS_PR_2000	985	760	225	0	225/985 (22.84%)	0/985 (0.00%)	985/1,000 (98.50%)	985/1,000 (98.50%)
Claude Haiku 5.5	mHS_HF_FU_4000	1992	1290	710	8	710/2,000 (35.50%)	8/2,000 (0.40%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Haiku 5.5	mHS_HF_PA_4000	1875	1721	279	125	279/2,000 (13.95%)	125/2,000 (6.25%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Haiku 5.5	mHS_HF_PR_4000	1966	1713	287	34	287/2,000 (14.35%)	34/2,000 (1.70%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Haiku 5.5	mHS_LF_FU_4000	1989	1076	913	2	913/1,989 (45.90%)	2/1,991 (0.10%)	1,989/2,000 (99.45%)	1,991/2,000 (99.55%)
Claude Haiku 5.5	mHS_LF_PA_4000	1991	895	1103	9	1,103/1,998 (55.21%)	9/2,000 (0.45%)	1,998/2,000 (99.90%)	2,000/2,000 (100.00%)
Claude Haiku 5.5	mHS_LF_PR_4000	1970	1057	926	15	926/1,983 (46.70%)	15/1,985 (0.76%)	1,983/2,000 (99.15%)	1,985/2,000 (99.25%)
Claude Haiku 5.5	mHS_NS_FU_4000	1973	1150	799	3	799/1,949 (41.00%)	3/1,976 (0.15%)	1,949/2,000 (97.45%)	1,976/2,000 (98.80%)
Claude Haiku 5.5	mHS_NS_PA_4000	1998	794	1195	2	1,195/1,989 (60.08%)	2/2,000 (0.10%)	1,989/2,000 (99.45%)	2,000/2,000 (100.00%)
Claude Haiku 5.5	mHS_NS_PR_4000	1983	1062	877	1	877/1,939 (45.23%)	1/1,984 (0.05%)	1,939/2,000 (96.95%)	1,984/2,000 (99.20%)
Claude Sonnet 5.5	HS_HF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_HF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_HF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_LF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)

Claude Sonnet 5.5	HS_LF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_LF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_NS_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_NS_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	HS_NS_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Claude Sonnet 5.5	mHS_HF_FU_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_HF_PA_4000	1990	2000	0	10	0/2,000 (0.00%)	10/2,000 (0.50%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_HF_PR_4000	1998	1999	0	2	0/1,999 (0.00%)	2/2,000 (0.10%)	1,999/2,000 (99.95%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_LF_FU_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_LF_PA_4000	1997	2000	0	3	0/2,000 (0.00%)	3/2,000 (0.15%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_LF_PR_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Claude Sonnet 5.5	mHS_NS_FU_4000	1999	2000	0	0	0/2,000 (0.00%)	0/1,999 (0.00%)	2,000/2,000 (100.00%)	1,999/2,000 (99.95%)
Claude Sonnet 5.5	mHS_NS_PA_4000	1999	2000	0	0	0/2,000 (0.00%)	0/1,999 (0.00%)	2,000/2,000 (100.00%)	1,999/2,000 (99.95%)
Claude Sonnet 5.5	mHS_NS_PR_4000	1991	2000	0	7	0/2,000 (0.00%)	7/1,998 (0.35%)	2,000/2,000 (100.00%)	1,998/2,000 (99.90%)
Gemini 3.1 Flash-Lite Flex	HS_HF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_HF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_HF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_LF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_LF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_LF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_NS_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_NS_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	HS_NS_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_HF_FU_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_HF_PA_4000	1961	2000	0	39	0/2,000 (0.00%)	39/2,000 (1.95%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_HF_PR_4000	1997	2000	0	3	0/2,000 (0.00%)	3/2,000 (0.15%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_LF_FU_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_LF_PA_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_LF_PR_4000	1999	2000	0	1	0/2,000 (0.00%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_NS_FU_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_NS_PA_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.1 Flash-Lite Flex	mHS_NS_PR_4000	2000	2000	0	0	0/2,000 (0.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_HF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_HF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_HF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)

Gemini 3.5 Flash-Lite Flex	HS_LF_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_LF_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_LF_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_NS_FU_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_NS_PA_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	HS_NS_PR_2000	1000	1000	0	0	0/1,000 (0.00%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_HF_FU_4000	1993	1948	52	7	52/2,000 (2.60%)	7/2,000 (0.35%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_HF_PA_4000	1754	1939	61	246	61/2,000 (3.05%)	246/2,000 (12.30%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_HF_PR_4000	1898	1945	55	102	55/2,000 (2.75%)	102/2,000 (5.10%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_LF_FU_4000	2000	1900	100	0	100/2,000 (5.00%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_LF_PA_4000	1992	1841	159	8	159/2,000 (7.95%)	8/2,000 (0.40%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_LF_PR_4000	1999	1858	142	1	142/2,000 (7.10%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_NS_FU_4000	2000	1893	107	0	107/2,000 (5.35%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_NS_PA_4000	1999	1841	159	1	159/2,000 (7.95%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS_NS_PR_4000	2000	1848	152	0	152/2,000 (7.60%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	HS_HF_FU_2000	990	995	5	10	5/1,000 (0.50%)	10/1,000 (1.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_HF_PA_2000	991	959	41	9	41/1,000 (4.10%)	9/1,000 (0.90%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_HF_PR_2000	990	988	12	10	12/1,000 (1.20%)	10/1,000 (1.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_LF_FU_2000	968	775	225	32	225/1,000 (22.50%)	32/1,000 (3.20%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_LF_PA_2000	898	724	276	102	276/1,000 (27.60%)	102/1,000 (10.20%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_LF_PR_2000	929	814	186	71	186/1,000 (18.60%)	71/1,000 (7.10%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_NS_FU_2000	870	995	5	130	5/1,000 (0.50%)	130/1,000 (13.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	HS_NS_PA_2000	864	974	26	135	26/1,000 (2.60%)	135/999 (13.51%)	1,000/1,000 (100.00%)	999/1,000 (99.90%)
GPT-4o mini	HS_NS_PR_2000	906	995	5	94	5/1,000 (0.50%)	94/1,000 (9.40%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-4o mini	mHS_HF_FU_4000	779	1996	4	1221	4/2,000 (0.20%)	1,221/2,000 (61.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_HF_PA_4000	550	1997	3	1450	3/2,000 (0.15%)	1,450/2,000 (72.50%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_HF_PR_4000	685	1995	5	1315	5/2,000 (0.25%)	1,315/2,000 (65.75%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_LF_FU_4000	984	1847	153	1016	153/2,000 (7.65%)	1,016/2,000 (50.80%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_LF_PA_4000	618	1969	31	1382	31/2,000 (1.55%)	1,382/2,000 (69.10%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_LF_PR_4000	879	1952	48	1121	48/2,000 (2.40%)	1,121/2,000 (56.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_NS_FU_4000	902	1995	5	1098	5/2,000 (0.25%)	1,098/2,000 (54.90%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_NS_PA_4000	654	1998	2	1346	2/2,000 (0.10%)	1,346/2,000 (67.30%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-4o mini	mHS_NS_PR_4000	812	1997	3	1188	3/2,000 (0.15%)	1,188/2,000 (59.40%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	HS_HF_FU_2000	1000	989	11	0	11/1,000 (1.10%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_HF_PA_2000	1000	994	6	0	6/1,000 (0.60%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)

GPT-6 Luna Flex	HS_HF_PR_2000	999	997	3	1	3/1,000 (0.30%)	1/1,000 (0.10%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_LF_FU_2000	998	318	682	2	682/1,000 (68.20%)	2/1,000 (0.20%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_LF_PA_2000	1000	485	515	0	515/1,000 (51.50%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_LF_PR_2000	999	651	349	1	349/1,000 (34.90%)	1/1,000 (0.10%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_NS_FU_2000	1000	949	51	0	51/1,000 (5.10%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_NS_PA_2000	1000	989	11	0	11/1,000 (1.10%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	HS_NS_PR_2000	1000	986	14	0	14/1,000 (1.40%)	0/1,000 (0.00%)	1,000/1,000 (100.00%)	1,000/1,000 (100.00%)
GPT-6 Luna Flex	mHS_HF_FU_4000	2000	1282	718	0	718/2,000 (35.90%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_HF_PA_4000	2000	1373	627	0	627/2,000 (31.35%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_HF_PR_4000	2000	1358	642	0	642/2,000 (32.10%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_LF_FU_4000	2000	843	1157	0	1,157/2,000 (57.85%)	0/2,000 (0.00%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_LF_PA_4000	1999	983	1017	1	1,017/2,000 (50.85%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_LF_PR_4000	1999	991	1009	1	1,009/2,000 (50.45%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_NS_FU_4000	1991	1554	446	9	446/2,000 (22.30%)	9/2,000 (0.45%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_NS_PA_4000	1999	1354	646	1	646/2,000 (32.30%)	1/2,000 (0.05%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)
GPT-6 Luna Flex	mHS_NS_PR_4000	1998	1414	586	2	586/2,000 (29.30%)	2/2,000 (0.10%)	2,000/2,000 (100.00%)	2,000/2,000 (100.00%)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FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
HS_HF_FU_2000	0.5% 5/1,000	1.1% 11/1,000	71.9% 719/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_HF_PA_2000	4.1% 41/1,000	0.6% 6/1,000	44.2% 442/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_HF_PR_2000	1.2% 12/1,000	0.3% 3/1,000	32.9% 329/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_FU_2000	22.5% 225/1,000	68.2% 682/1,000	89.5% 890/994	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_PA_2000	27.6% 276/1,000	51.5% 515/1,000	78.8% 788/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_PR_2000	18.6% 186/1,000	34.9% 349/1,000	67.1% 667/994	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_FU_2000	0.5% 5/1,000	5.1% 51/1,000	65.4% 627/958	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_PA_2000	2.6% 26/1,000	1.1% 11/1,000	46.1% 461/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_PR_2000	0.5% 5/1,000	1.4% 14/1,000	22.8% 225/985	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
mHS_HF_FU_4000	0.2% 4/2,000	35.9% 718/2,000	35.5% 710/2,000	0.0% 0/2,000	0.0% 0/2,000	2.6% 52/2,000
mHS_HF_PA_4000	0.1% 3/2,000	31.4% 627/2,000	14.0% 279/2,000	0.0% 0/2,000	0.0% 0/2,000	3.0% 61/2,000
mHS_HF_PR_4000	0.2% 5/2,000	32.1% 642/2,000	14.3% 287/2,000	0.0% 0/1,999	0.0% 0/2,000	2.8% 55/2,000
mHS_LF_FU_4000	7.6% 153/2,000	57.9% 1,157/2,000	45.9% 913/1,989	0.0% 0/2,000	0.0% 0/2,000	5.0% 100/2,000
mHS_LF_PA_4000	1.6% 31/2,000	50.8% 1,017/2,000	55.2% 1,103/1,998	0.0% 0/2,000	0.0% 0/2,000	8.0% 159/2,000
mHS_LF_PR_4000	2.4% 48/2,000	50.4% 1,009/2,000	46.7% 926/1,983	0.0% 0/2,000	0.0% 0/2,000	7.1% 142/2,000
mHS_NS_FU_4000	0.2% 5/2,000	22.3% 446/2,000	41.0% 799/1,949	0.0% 0/2,000	0.0% 0/2,000	5.3% 107/2,000
mHS_NS_PA_4000	0.1% 2/2,000	32.3% 646/2,000	60.1% 1,195/1,989	0.0% 0/2,000	0.0% 0/2,000	8.0% 159/2,000
mHS_NS_PR_4000	0.1% 3/2,000	29.3% 586/2,000	45.2% 877/1,939	0.0% 0/2,000	0.0% 0/2,000	7.6% 152/2,000

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
HS_HF_FU_2000	1.0% 10/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_HF_PA_2000	0.9% 9/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_HF_PR_2000	1.0% 10/1,000	0.1% 1/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_FU_2000	3.2% 32/1,000	0.2% 2/1,000	0.0% 0/994	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_PA_2000	10.2% 102/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_LF_PR_2000	7.1% 71/1,000	0.1% 1/1,000	0.0% 0/996	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_FU_2000	13.0% 130/1,000	0.0% 0/1,000	0.0% 0/959	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_PA_2000	13.5% 135/999	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
HS_NS_PR_2000	9.4% 94/1,000	0.0% 0/1,000	0.0% 0/985	0.0% 0/1,000	0.0% 0/1,000	0.0% 0/1,000
mHS_HF_FU_4000	61.1% 1,221/2,000	0.0% 0/2,000	0.4% 8/2,000	0.0% 0/2,000	0.0% 0/2,000	0.4% 7/2,000
mHS_HF_PA_4000	72.5% 1,450/2,000	0.0% 0/2,000	6.2% 125/2,000	0.5% 10/2,000	1.9% 39/2,000	12.3% 246/2,000
mHS_HF_PR_4000	65.8% 1,315/2,000	0.0% 0/2,000	1.7% 34/2,000	0.1% 2/2,000	0.1% 3/2,000	5.1% 102/2,000
mHS_LF_FU_4000	50.8% 1,016/2,000	0.0% 0/2,000	0.1% 2/1,991	0.0% 0/2,000	0.0% 0/2,000	0.0% 0/2,000
mHS_LF_PA_4000	69.1% 1,382/2,000	0.1% 1/2,000	0.4% 9/2,000	0.1% 3/2,000	0.0% 0/2,000	0.4% 8/2,000
mHS_LF_PR_4000	56.0% 1,121/2,000	0.1% 1/2,000	0.8% 15/1,985	0.0% 0/2,000	0.1% 1/2,000	0.1% 1/2,000
mHS_NS_FU_4000	54.9% 1,098/2,000	0.4% 9/2,000	0.2% 3/1,976	0.0% 0/1,999	0.0% 0/2,000	0.0% 0/2,000
mHS_NS_PA_4000	67.3% 1,346/2,000	0.1% 1/2,000	0.1% 2/2,000	0.0% 0/1,999	0.0% 0/2,000	0.1% 1/2,000
mHS_NS_PR_4000	59.4% 1,188/2,000	0.1% 2/2,000	0.1% 1/1,984	0.4% 7/1,998	0.0% 0/2,000	0.0% 0/2,000

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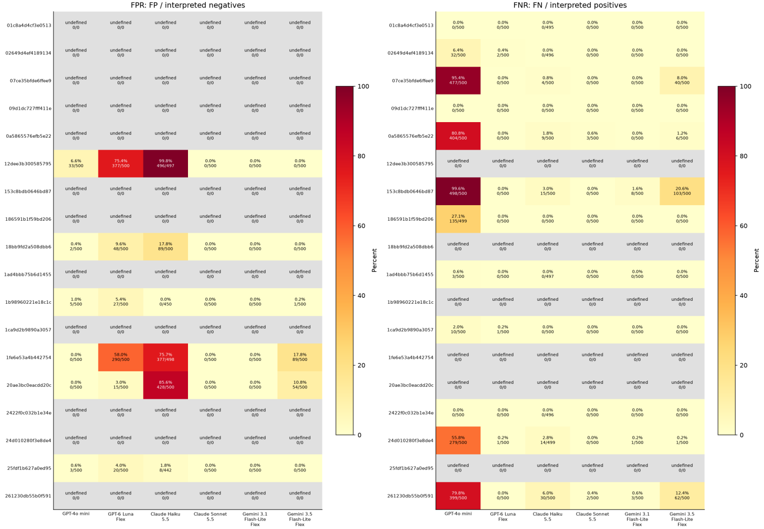

FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
01c8a4d4cf3e0513	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
02649d4ef4189134	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
07ce35bfde6ffee9	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
09d1dc727fff411e	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
0a5865576efb5e22	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
12dee3b300585795	6.6% 33/500	75.4% 377/500	99.8% 496/497	0.0% 0/500	0.0% 0/500	0.0% 0/500
153c8bdb0646bd87	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
186591b1f59bd206	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
18bb9fd2a508dbb6	0.4% 2/500	9.6% 48/500	17.8% 89/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
1ad4bbb75b6d1455	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
1b98960221e18c1c	1.0% 5/500	5.4% 27/500	0.0% 0/450	0.0% 0/500	0.0% 0/500	0.2% 1/500
1ca9d2b9890a3057	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
1fe6e53a4b442754	0.0% 0/500	58.0% 290/500	75.7% 377/498	0.0% 0/500	0.0% 0/500	17.8% 89/500
20ae3bc0eacdd20c	0.0% 0/500	3.0% 15/500	85.6% 428/500	0.0% 0/500	0.0% 0/500	10.8% 54/500
2422f0c032b1e34e	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
24d010280f3e8de4	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
25fdf1b627a0ed95	0.6% 3/500	4.0% 20/500	1.8% 8/442	0.0% 0/500	0.0% 0/500	0.0% 0/500
261230db55b0f591	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
01c8a4d4cf3e0513	0.0% 0/500	0.0% 0/500	0.0% 0/495	0.0% 0/500	0.0% 0/500	0.0% 0/500
02649d4ef4189134	6.4% 32/500	0.4% 2/500	0.0% 0/496	0.0% 0/500	0.0% 0/500	0.0% 0/500
07ce35bfde6ffee9	95.4% 477/500	0.0% 0/500	0.8% 4/500	0.0% 0/500	0.0% 0/500	8.0% 40/500
09d1dc727fff411e	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
0a5865576efb5e22	80.8% 404/500	0.0% 0/500	1.8% 9/500	0.6% 3/500	0.0% 0/500	1.2% 6/500
12dee3b300585795	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
153c8bdb0646bd87	99.6% 498/500	0.0% 0/500	3.0% 15/500	0.0% 0/500	1.6% 8/500	20.6% 103/500
186591b1f59bd206	27.1% 135/499	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
18bb9fd2a508dbb6	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
1ad4bbb75b6d1455	0.6% 3/500	0.0% 0/500	0.0% 0/497	0.0% 0/500	0.0% 0/500	0.0% 0/500
1b98960221e18c1c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
1ca9d2b9890a3057	2.0% 10/500	0.2% 1/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
1fe6e53a4b442754	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
20ae3bc0eacdd20c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
2422f0c032b1e34e	0.0% 0/500	0.0% 0/500	0.0% 0/496	0.0% 0/500	0.0% 0/500	0.0% 0/500
24d010280f3e8de4	55.8% 279/500	0.2% 1/500	2.8% 14/499	0.0% 0/500	0.2% 1/500	0.2% 1/500
25fdf1b627a0ed95	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
261230db55b0f591	79.8% 399/500	0.0% 0/500	6.0% 30/500	0.4% 2/500	0.6% 3/500	12.4% 62/500

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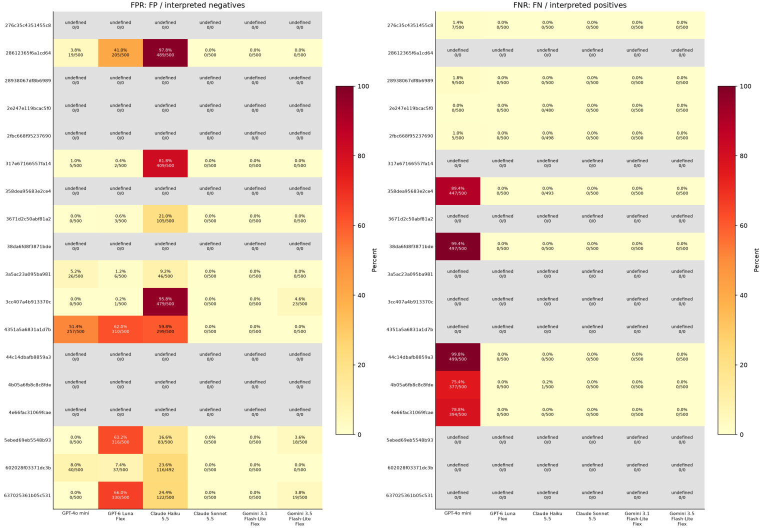
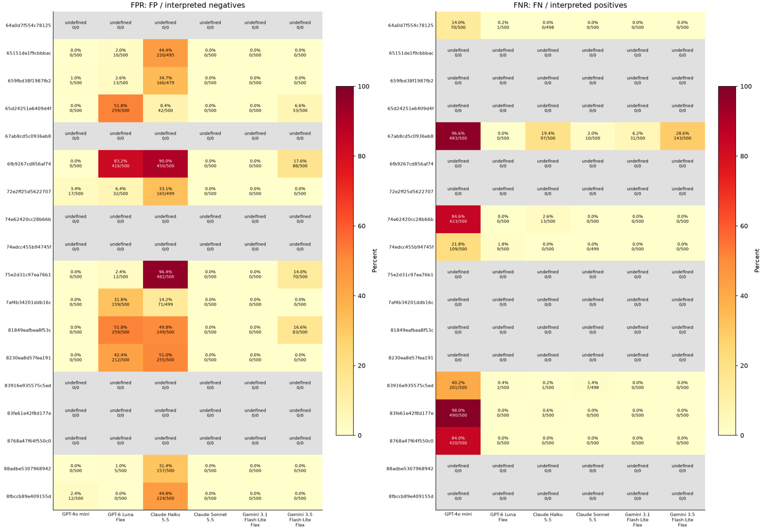


FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
276c35c4351455c8	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
28612365f6a1cd64	3.8% 19/500	41.0% 205/500	97.8% 489/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
28938067df8b6989	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
2e247e119bcac5f0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
2fbc668f95237690	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
317e67166557fa14	1.0% 5/500	0.4% 2/500	81.8% 409/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
358dea95683e2ce4	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
3671d2c50abf81a2	0.0% 0/500	0.6% 3/500	21.0% 105/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
38da6fd8f3871bde	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
3a5ac23a095ba981	5.2% 26/500	1.2% 6/500	9.2% 46/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
3cc407a4b913370c	0.0% 0/500	0.2% 1/500	95.8% 479/500	0.0% 0/500	0.0% 0/500	4.6% 23/500
4351a5a6831a1d7b	51.4% 257/500	62.0% 310/500	59.8% 299/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
44c14dbafb8859a3	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
4b05a6fb8c8c8fde	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
4e66fac31069fcae	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
5ebed69eb5548b93	0.0% 0/500	63.2% 316/500	16.6% 83/500	0.0% 0/500	0.0% 0/500	3.6% 18/500
602028f03371dc3b	8.0% 40/500	7.4% 37/500	23.6% 116/492	0.0% 0/500	0.0% 0/500	0.0% 0/500
637025361b05c531	0.0% 0/500	66.0% 330/500	24.4% 122/500	0.0% 0/500	0.0% 0/500	3.8% 19/500

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
276c35c4351455c8	1.4% 7/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
28612365f6a1cd64	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
28938067df8b6989	1.8% 9/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
2e247e119bcac5f0	0.0% 0/500	0.0% 0/500	0.0% 0/480	0.0% 0/500	0.0% 0/500	0.0% 0/500
2fbc668f95237690	1.0% 5/500	0.0% 0/500	0.0% 0/498	0.0% 0/500	0.0% 0/500	0.0% 0/500
317e67166557fa14	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
358dea95683e2ce4	89.4% 447/500	0.0% 0/500	0.0% 0/493	0.0% 0/500	0.0% 0/500	0.0% 0/500
3671d2c50abf81a2	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
38da6fd8f3871bde	99.4% 497/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
3a5ac23a095ba981	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
3cc407a4b913370c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
4351a5a6831a1d7b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
44c14dbafb8859a3	99.8% 499/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
4b05a6fb8c8c8fde	75.4% 377/500	0.0% 0/500	0.2% 1/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
4e66fac31069fcae	78.8% 394/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
5ebed69eb5548b93	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
602028f03371dc3b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
637025361b05c531	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0



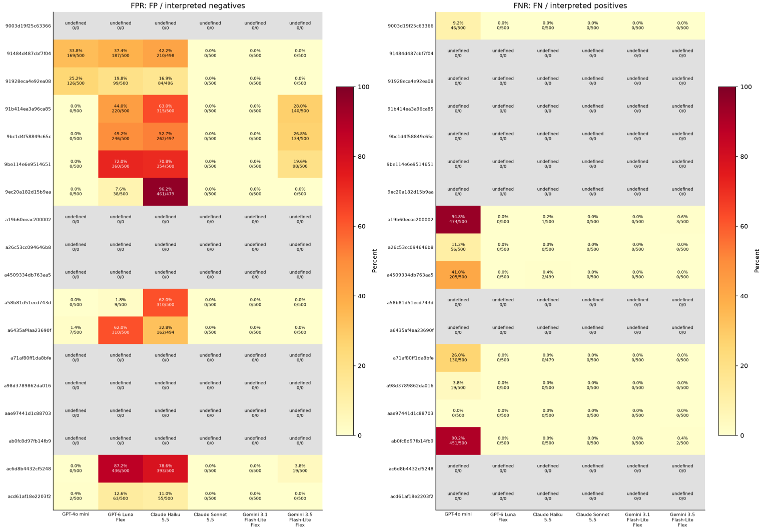
stratum: source-label error rates (gray = undefined)

FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
64a0d7f554c78125	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
65151de1f9cbbbac	0.0% 0/500	2.0% 10/500	44.4% 220/495	0.0% 0/500	0.0% 0/500	0.0% 0/500
659fbd38f1987fb2	1.0% 5/500	2.6% 13/500	34.7% 166/479	0.0% 0/500	0.0% 0/500	0.0% 0/500
65d24251e6409d4f	0.0% 0/500	51.8% 259/500	8.4% 42/500	0.0% 0/500	0.0% 0/500	6.6% 33/500
67ab8cd5c0936eb8	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
6fb9267cd856af74	0.0% 0/500	83.2% 416/500	90.0% 450/500	0.0% 0/500	0.0% 0/500	17.6% 88/500
72e2ff25d5622707	3.4% 17/500	6.4% 32/500	33.1% 165/499	0.0% 0/500	0.0% 0/500	0.0% 0/500
74e62420cc28b66b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
74edcc455b94745f	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
75e2d31c97ea76b1	0.0% 0/500	2.4% 12/500	96.4% 482/500	0.0% 0/500	0.0% 0/500	14.0% 70/500
7af4b34201ddb16c	0.0% 0/500	31.8% 159/500	14.2% 71/499	0.0% 0/500	0.0% 0/500	0.0% 0/500
81849eafbea8f53c	0.0% 0/500	51.8% 259/500	49.8% 249/500	0.0% 0/500	0.0% 0/500	16.6% 83/500
8230ea8d57fea191	0.0% 0/500	42.4% 212/500	51.0% 255/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
83916e935575c5ed	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
83fe61e42f8d177e	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
8768a47f64f550c0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
88adbe5307968942	0.0% 0/500	1.0% 5/500	31.4% 157/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
8fbccb89e409155d	2.4% 12/500	0.0% 0/500	44.8% 224/500	0.0% 0/500	0.0% 0/500	0.0% 0/500

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
64a0d7f554c78125	14.0% 70/500	0.2% 1/500	0.0% 0/498	0.0% 0/500	0.0% 0/500	0.0% 0/500
65151de1f9cbbbac	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
659fbd38f1987fb2	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
65d24251e6409d4f	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
67ab8cd5c0936eb8	96.6% 483/500	0.0% 0/500	19.4% 97/500	2.0% 10/500	6.2% 31/500	28.6% 143/500
6fb9267cd856af74	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
72e2ff25d5622707	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
74e62420cc28b66b	84.6% 423/500	0.0% 0/500	2.6% 13/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
74edcc455b94745f	21.8% 109/500	1.8% 9/500	0.0% 0/500	0.0% 0/499	0.0% 0/500	0.0% 0/500
75e2d31c97ea76b1	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
7af4b34201ddb16c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
81849eafbea8f53c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
8230ea8d57fea191	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
83916e935575c5ed	40.2% 201/500	0.4% 2/500	0.2% 1/500	1.4% 7/498	0.0% 0/500	0.0% 0/500
83fe61e42f8d177e	98.0% 490/500	0.0% 0/500	0.6% 3/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
8768a47f64f550c0	84.0% 420/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
88adbe5307968942	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
8fbccb89e409155d	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0



stratum: source-label error rates (gray = undefined)

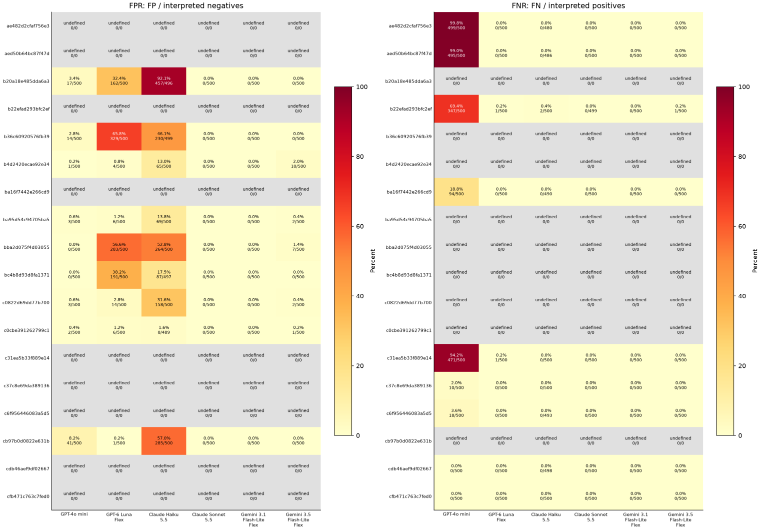
FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
9003d19f25c63366	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
91484d487cbf7f04	33.8% 169/500	37.4% 187/500	42.2% 210/498	0.0% 0/500	0.0% 0/500	0.0% 0/500
91928eca4e92ea08	25.2% 126/500	19.8% 99/500	16.9% 84/496	0.0% 0/500	0.0% 0/500	0.0% 0/500
91b414ea3a96ca85	0.0% 0/500	44.0% 220/500	63.0% 315/500	0.0% 0/500	0.0% 0/500	28.0% 140/500
9bc1d4f58849c65c	0.0% 0/500	49.2% 246/500	52.7% 262/497	0.0% 0/500	0.0% 0/500	26.8% 134/500
9be114e6e9514651	0.0% 0/500	72.0% 360/500	70.8% 354/500	0.0% 0/500	0.0% 0/500	19.6% 98/500
9ec20a182d15b9aa	0.0% 0/500	7.6% 38/500	96.2% 461/479	0.0% 0/500	0.0% 0/500	0.0% 0/500
a19b60eeac200002	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a26c53cc094646b8	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a4509334db763aa5	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a58b81d51ecd743d	0.0% 0/500	1.8% 9/500	62.0% 310/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
a6435af4aa23690f	1.4% 7/500	62.0% 310/500	32.8% 162/494	0.0% 0/500	0.0% 0/500	0.0% 0/500
a71af80ff1da8bfe	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a98d3789862da016	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
aae97441d1c88703	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ab0fc8d97fb14fb9	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ac6d8b4432cf5248	0.0% 0/500	87.2% 436/500	78.6% 393/500	0.0% 0/500	0.0% 0/500	3.8% 19/500
acd61af18e2203f2	0.4% 2/500	12.6% 63/500	11.0% 55/500	0.0% 0/500	0.0% 0/500	0.0% 0/500

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
9003d19f25c63366	9.2% 46/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
91484d487cbf7f04	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
91928eca4e92ea08	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
91b414ea3a96ca85	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
9bc1d4f58849c65c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
9be114e6e9514651	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
9ec20a182d15b9aa	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a19b60eeac200002	94.8% 474/500	0.0% 0/500	0.2% 1/500	0.0% 0/500	0.0% 0/500	0.6% 3/500
a26c53cc094646b8	11.2% 56/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
a4509334db763aa5	41.0% 205/500	0.0% 0/500	0.4% 2/499	0.0% 0/500	0.0% 0/500	0.0% 0/500
a58b81d51ecd743d	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a6435af4aa23690f	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
a71af80ff1da8bfe	26.0% 130/500	0.0% 0/500	0.0% 0/479	0.0% 0/500	0.0% 0/500	0.0% 0/500
a98d3789862da016	3.8% 19/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
aae97441d1c88703	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
ab0fc8d97fb14fb9	90.2% 451/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.4% 2/500
ac6d8b4432cf5248	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
acd61af18e2203f2	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0

stratum: source-label error rates (gray = undefined)



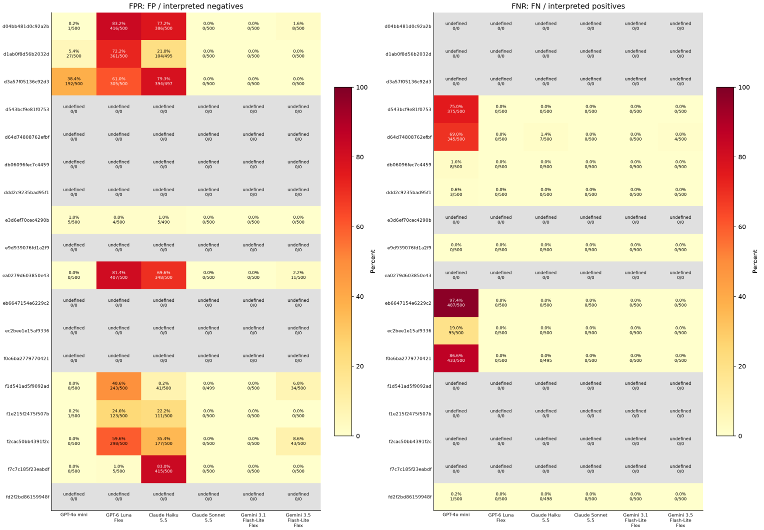
FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
ae482d2cfaf756e3	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
aed50b64bc87f47d	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
b20a18e485dda6a3	3.4% 17/500	32.4% 162/500	92.1% 457/496	0.0% 0/500	0.0% 0/500	0.0% 0/500
b22efad293bfc2ef	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
b36c60920576fb39	2.8% 14/500	65.8% 329/500	46.1% 230/499	0.0% 0/500	0.0% 0/500	0.0% 0/500
b4d2420ecae92e34	0.2% 1/500	0.8% 4/500	13.0% 65/500	0.0% 0/500	0.0% 0/500	2.0% 10/500
ba16f7442e266cd9	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ba95d54c94705ba5	0.6% 3/500	1.2% 6/500	13.8% 69/500	0.0% 0/500	0.0% 0/500	0.4% 2/500
bba2d075f4d03055	0.0% 0/500	56.6% 283/500	52.8% 264/500	0.0% 0/500	0.0% 0/500	1.4% 7/500
bc4b8d93d8fa1371	0.0% 0/500	38.2% 191/500	17.5% 87/497	0.0% 0/500	0.0% 0/500	0.0% 0/500
c0822d69dd77b700	0.6% 3/500	2.8% 14/500	31.6% 158/500	0.0% 0/500	0.0% 0/500	0.4% 2/500
c0cbe391262799c1	0.4% 2/500	1.2% 6/500	1.6% 8/489	0.0% 0/500	0.0% 0/500	0.2% 1/500
c31ea5b33f889e14	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
c37c8e69da389136	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
c6f956446083a5d5	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
cb97b0d0822e631b	8.2% 41/500	0.2% 1/500	57.0% 285/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
cdb46aef9df02667	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
cfb471c763c7fed0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
ae482d2cfaf756e3	99.8% 499/500	0.0% 0/500	0.0% 0/480	0.0% 0/500	0.0% 0/500	0.0% 0/500
aed50b64bc87f47d	99.0% 495/500	0.0% 0/500	0.0% 0/486	0.0% 0/500	0.0% 0/500	0.0% 0/500
b20a18e485dda6a3	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
b22efad293bfc2ef	69.4% 347/500	0.2% 1/500	0.4% 2/500	0.0% 0/499	0.0% 0/500	0.2% 1/500
b36c60920576fb39	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
b4d2420ecae92e34	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ba16f7442e266cd9	18.8% 94/500	0.0% 0/500	0.0% 0/490	0.0% 0/500	0.0% 0/500	0.0% 0/500
ba95d54c94705ba5	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
bba2d075f4d03055	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
bc4b8d93d8fa1371	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
c0822d69dd77b700	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
c0cbe391262799c1	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
c31ea5b33f889e14	94.2% 471/500	0.2% 1/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
c37c8e69da389136	2.0% 10/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
c6f956446083a5d5	3.6% 18/500	0.0% 0/500	0.0% 0/493	0.0% 0/500	0.0% 0/500	0.0% 0/500
cb97b0d0822e631b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
cdb46aef9df02667	0.0% 0/500	0.0% 0/500	0.0% 0/498	0.0% 0/500	0.0% 0/500	0.0% 0/500
cfb471c763c7fed0	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500

stratum: source-label error rates (gray = undefined)



FPR: FP / interpreted negatives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
d04bb481d0c92a2b	0.2% 1/500	83.2% 416/500	77.2% 386/500	0.0% 0/500	0.0% 0/500	1.6% 8/500
d1ab0f8d56b2032d	5.4% 27/500	72.2% 361/500	21.0% 104/495	0.0% 0/500	0.0% 0/500	0.0% 0/500
d3a57f05136c92d3	38.4% 192/500	61.0% 305/500	79.3% 394/497	0.0% 0/500	0.0% 0/500	0.0% 0/500
d543bcf9e81f0753	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
d64d74808762efbf	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
db06096fec7c4459	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ddd2c9235bad95f1	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
e3d6ef70cec4290b	1.0% 5/500	0.8% 4/500	1.0% 5/490	0.0% 0/500	0.0% 0/500	0.0% 0/500
e9d939076fd1a2f9	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ea0279d603850e43	0.0% 0/500	81.4% 407/500	69.6% 348/500	0.0% 0/500	0.0% 0/500	2.2% 11/500
eb6647154e6229c2	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
ec2bee1e15af9336	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
f0e6ba2779770421	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
f1d541ad5f9092ad	0.0% 0/500	48.6% 243/500	8.2% 41/500	0.0% 0/499	0.0% 0/500	6.8% 34/500
f1e215f2475f507b	0.2% 1/500	24.6% 123/500	22.2% 111/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
f2cac50bb4391f2c	0.0% 0/500	59.6% 298/500	35.4% 177/500	0.0% 0/500	0.0% 0/500	8.6% 43/500
f7c7c185f23eabdf	0.0% 0/500	1.0% 5/500	83.0% 415/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
fd2f2bd86159948f	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0

FNR: FN / interpreted positives

	GPT-4o mini	GPT-6 Luna Flex	Claude Haiku 5.5	Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex
d04bb481d0c92a2b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
d1ab0f8d56b2032d	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
d3a57f05136c92d3	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
d543bcf9e81f0753	75.0% 375/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
d64d74808762efbf	69.0% 345/500	0.0% 0/500	1.4% 7/500	0.0% 0/500	0.0% 0/500	0.8% 4/500
db06096fec7c4459	1.6% 8/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
ddd2c9235bad95f1	0.6% 3/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
e3d6ef70cec4290b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
e9d939076fd1a2f9	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
ea0279d603850e43	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
eb6647154e6229c2	97.4% 487/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
ec2bee1e15af9336	19.0% 95/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500	0.0% 0/500
f0e6ba2779770421	86.6% 433/500	0.0% 0/500	0.0% 0/495	0.0% 0/500	0.0% 0/500	0.0% 0/500
f1d541ad5f9092ad	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
f1e215f2475f507b	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
f2cac50bb4391f2c	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
f7c7c185f23eabdf	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0	undefined 0/0
fd2f2bd86159948f	0.2% 1/500	0.0% 0/500	0.0% 0/498	0.0% 0/500	0.0% 0/500	0.0% 0/500

Every heatmap uses the same 0-100% scale and prints its numerator and denominator. Gray cells are undefined, not zero. Stratum IDs map to tables/stratum_definitions.csv. The condition CSVs retain all counts and class coverage values. The separate explorer recomputes rates from counts when filters are combined.

Do models disagree on the same cases?

Each pair contains the same 54,000 UIDs. Agreement between answers and one-model-only source matches use only cases where both models gave interpretable answers. The excluded count remains visible. No missing answer is treated as a disagreement.

Model A	Model B	Same answer / both interpreted	Only A matches	Only B matches	Both match	Both disagree	Not both interpreted
GPT-4o mini	GPT-6 Luna Flex	33,628/53,999 (62.28%)	8057	12314	33177	451	1
GPT-4o mini	Claude Haiku 5.5	29,911/53,647 (55.76%)	11732	12004	29207	704	353
GPT-4o mini	Claude Sonnet 5.5	41,243/53,994 (76.38%)	4	12747	41225	18	6
GPT-4o mini	Gemini 3.1 Flash-Lite Flex	41,275/53,999 (76.44%)	1	12723	41233	42	1
GPT-4o mini	Gemini 3.5 Flash-Lite Flex	40,584/53,999 (75.16%)	1001	12414	40233	351	1
GPT-6 Luna Flex	Claude Haiku 5.5	41,933/53,648 (78.16%)	7834	3881	37331	4602	352
GPT-6 Luna Flex	Claude Sonnet 5.5	45,465/53,995 (84.20%)	22	8508	45465	0	5
GPT-6 Luna Flex	Gemini 3.1 Flash-Lite Flex	45,449/54,000 (84.16%)	43	8508	45449	0	0
GPT-6 Luna Flex	Gemini 3.5 Flash-Lite Flex	45,110/54,000 (83.54%)	867	8023	44625	485	0
Claude Haiku 5.5	Claude Sonnet 5.5	41,191/53,643 (76.79%)	19	12433	41188	3	357
Claude Haiku 5.5	Gemini 3.1 Flash-Lite Flex	41,195/53,648 (76.79%)	30	12423	41182	13	352
Claude Haiku 5.5	Gemini 3.5 Flash-Lite Flex	41,236/53,648 (76.86%)	664	11748	40548	688	352
Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	53,930/53,995 (99.88%)	43	22	53930	0	5
Claude Sonnet 5.5	Gemini 3.5 Flash-Lite Flex	52,631/53,995 (97.47%)	1347	17	52626	5	5
Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex	52,651/54,000 (97.50%)	1329	20	52628	23	0

Cases that make the results concrete

Examples are selected by the smallest SHA-256 hash of the UID for each family and category, without repeating a case. Family order is HS then mHS; category order is different answers, content filter, then interpretable non-strict format. These examples illustrate outcomes; they do not estimate prevalence. The HTML explorer exposes every case.

HS: different answers

UID: HS_NS_FU_2000:955:ecd5d089ff37fde8. Dataset: HS_NS_FU_2000; original row 955. Source label: invalid; expected answer: no.

```
If the preggar will glash the ploodnum, then the probeller will trelf the blorrack; if the probeller will trelf the blorrack, then the bruggant will frike the drambet; therefore if the bruggant will frike the drambet, then the preggar will glash the ploodnum?
```

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	yes	disagreement	True
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	no	match	True

Claude Haiku 5.5 — original response

```
yes
```

Claude Sonnet 5.5 — original response

```
no
```

Gemini 3.1 Flash-Lite Flex — original response no

Gemini 3.5 Flash-Lite Flex — original response

no

GPT-4o mini — original response

no

GPT-6 Luna Flex — original response

no

HS: content filter

UID: HS_NS_PR_2000:37:7f42d42c0023eb28. Dataset: HS_NS_PR_2000; original row 37. Source label: valid; expected answer: yes.

If the preggar gresks the prebart, then the brettent glashes the flendop; if the brettent glashes the flendop, then the probeller frikes the ploodnum; therefore if the preggar gresks the prebart, then the probeller frikes the ploodnum?

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	not interpreted	content_filter	False
Claude Sonnet 5.5	yes	match	True
Gemini 3.1 Flash-Lite Flex	yes	match	True
Gemini 3.5 Flash-Lite Flex	yes	match	True
GPT-4o mini	yes	match	True
GPT-6 Luna Flex	yes	match	True

Claude Haiku 5.5 — original response

This request triggered restrictions on content related to the development of high-yield explosives and was blocked under Anthropic's Usage Policy. To learn more, provide feedback, or request an exemption based on how you use Claude, visit our help center: https://support.claude.com/ API integrators: you can reduce refusals for your users by configuring a fallback model — see https://platform.claude.com/docs/en/build-with-claude/refusals-and-fallback

Claude Sonnet 5.5 — original response

yes

Gemini 3.1 Flash-Lite Flex — original response

yes

Gemini 3.5 Flash-Lite Flex — original response

yes

GPT-4o mini — original response

yes

GPT-6 Luna Flex — original response

yes

HS: interpretable, non-strict format

UID: HS_HF_FU_2000:1602:787e8fed73f8e1dc. Dataset: HS_HF_FU_2000; original row 1602. Source label: invalid; expected answer: no.

If the student will hate the computer, then the family will see the college; if the doctor will win the game, then the student will hate the computer; therefore if the family will see the college, then the doctor will win the game?

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	no	match	True
Claude Sonnet 5.5	no	match	False
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	no	match	True

Claude Haiku 5.5 — original response

no

Claude Sonnet 5.5 — original response

Let H = the student will hate the computer, F = the family will see the college, D = the doctor will win the game.

The premises are H → F and D → H. Chaining them gives D → F.

The conclusion asks for F → D, which is the converse of D → F. The premises do not support it. For example, if F is true while D and H are both false, both premises hold and the conclusion fails.

No

Gemini 3.1 Flash-Lite Flex — original response

no

Gemini 3.5 Flash-Lite Flex — original response

no

GPT-4o mini — original response

no

GPT-6 Luna Flex — original response

no

mHS: different answers

UID: mHS_HF_PR_4000:1953:d7b258905cf06b38. Dataset: mHS_HF_PR_4000; original row 1953. Source label: invalid; expected answer: no.

From the inference 'the woman pays the sugar, therefore the girl hates the school', and the inference 'the girl hates the school, therefore the officer finds the college', does the inference 'the officer finds the college, therefore the woman pays the sugar' follow?

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	no	match	True
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	yes	disagreement	True

Claude Haiku 5.5 — original response

no

Claude Sonnet 5.5 — original response

no

Gemini 3.1 Flash-Lite Flex — original response

no

Gemini 3.5 Flash-Lite Flex — original response

no

GPT-4o mini — original response

no

GPT-6 Luna Flex — original response

yes

mHS: content filter

UID: mHS_NS_FU_4000:1062:cbe3c3e2be728394. Dataset: mHS_NS_FU_4000; original row 1062. Source label: invalid; expected answer: no.

Assuming that the bruggant will trelf the drambet, we can infer that the preggar will prant the ploodnum. Assuming that the preggar will prant the ploodnum, we can infer that the probeller will glash the prebart. Thus, assuming that the probeller will glash the prebart, can we infer that the bruggant will trelf the drambet?

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	not interpreted	content_filter	False
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True

Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	yes	disagreement	True

Claude Haiku 5.5 — original response

This request triggered restrictions on content related to the development of high-yield explosives and was blocked under Anthropic's Usage Policy. To learn more, provide feedback, or request an exemption based on how you use Claude, visit our help center: https://support.claude.com/ API integrators: you can reduce refusals for your users by configuring a fallback model — see https://platform.claude.com/docs/en/build-with-claude/refusals-and-fallback

Claude Sonnet 5.5 — original response

no

Gemini 3.1 Flash-Lite Flex — original response

no

Gemini 3.5 Flash-Lite Flex — original response

no

GPT-4o mini — original response

no

GPT-6 Luna Flex — original response

yes

mHS: interpretable, non-strict format

UID: mHS_HF_PR_4000:153:99c4ec3b9f63ae06. Dataset: mHS_HF_PR_4000; original row 153. Source label: valid; expected answer: yes.

Assuming that the family pays the computer, we can infer that the man loses the game. Assuming that the man loses the game, we can infer that the woman hates the school. Thus, assuming that the family pays the computer, can we infer that the woman hates the school?

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	yes	match	True
Claude Sonnet 5.5	yes	match	True
Gemini 3.1 Flash-Lite Flex	yes	match	True
Gemini 3.5 Flash-Lite Flex	yes	match	True
GPT-4o mini	yes	match	False
GPT-6 Luna Flex	yes	match	True

Claude Haiku 5.5 — original response

yes

Claude Sonnet 5.5 — original response

yes

Gemini 3.1 Flash-Lite Flex — original response

yes

Gemini 3.5 Flash-Lite Flex — original response

Yes

GPT-4o mini — original response

Yes.

GPT-6 Luna Flex — original response

yes

What did the run cost?

These are costs reported by OpenRouter, not an independent invoice audit. Costs are deduplicated by generation ID. A captured response cost takes precedence over a later lookup; conflicting observations remain in the database and conflict table. Zero is retained as zero. Missing costs remain unknown.

Phase	Model	Generation role	IDs with cost / all IDs	Unknown-cost IDs	Known USD

corpus	Claude Haiku 5.5	final	53648/54000	352	0.789825
corpus	Claude Sonnet 5.5	final	54000/54000	0	26.645994
corpus	Gemini 3.1 Flash-Lite Flex	final	54000/54000	0	0.615776
corpus	Gemini 3.5 Flash-Lite Flex	final	54000/54000	0	0.757811
corpus	GPT-4o mini	final	54000/54000	0	0.804191
corpus	GPT-6 Luna Flex	additional	0/426	426	unknown
corpus	GPT-6 Luna Flex	final	54000/54000	0	0.321318
correction_preflight	Claude Haiku 5.5	final	1/1	0	0.000011
correction_preflight	Claude Sonnet 5.5	final	1/1	0	0.000218
correction_preflight	Gemini 3.1 Flash-Lite Flex	final	1/1	0	0.000008
correction_preflight	Gemini 3.5 Flash-Lite Flex	final	1/1	0	0.000010
correction_preflight	GPT-4o mini	final	1/1	0	0.000011
correction_preflight	GPT-6 Luna Flex	final	1/1	0	0.000005
preflight	Claude Haiku 5.5	final	1/1	0	0.000012
preflight	Claude Sonnet 5.5	final	1/1	0	0.000234
preflight	Gemini 3.1 Flash-Lite Flex	final	1/1	0	0.000009
preflight	Gemini 3.5 Flash-Lite Flex	final	1/1	0	0.000011
preflight	GPT-4o mini	final	1/1	0	0.000012
preflight	GPT-6 Luna Flex	final	1/1	0	0.000005
superseded	Claude Haiku 5.5	historical replaced calls	612/612	0	0.007434
superseded	Claude Sonnet 5.5	historical replaced calls	612/612	0	0.231660
superseded	Gemini 3.1 Flash-Lite Flex	historical replaced calls	612/612	0	0.005712
superseded	Gemini 3.5 Flash-Lite Flex	historical replaced calls	612/612	0	0.007070
superseded	GPT-4o mini	historical replaced calls	612/612	0	0.007550
superseded	GPT-6 Luna Flex	historical replaced calls	612/613	1	0.003121

Across both executions, including superseded calls and both preflights, known reported cost is USD 30.198008. There are 779 generation IDs with unknown cost and 0 conflicting cost observations. The superseded phase is historical cost only and is excluded from current outcome measures.

Final generations have an explicit final response ID. Additional generations are other identified attempts for a sample with a final ID. Unresolved generations belong to samples without an explicit final ID; their role is not guessed. Attempts without IDs cannot be assigned a deduplicated cost.

Phase	Model	HTTP attempts	Failed attempts	Attempts without ID	Without end timestamp
corpus	Claude Haiku 5.5	54000	0	0	0
corpus	Claude Sonnet 5.5	54000	0	0	0
corpus	Gemini 3.1 Flash-Lite Flex	54001	1	1	0
corpus	Gemini 3.5 Flash-Lite Flex	54000	0	0	0
corpus	GPT-4o mini	54000	0	0	0
corpus	GPT-6 Luna Flex	54426	426	0	0
correction_preflight	Claude Haiku 5.5	1	0	0	0
correction_preflight	Claude Sonnet 5.5	1	0	0	0
correction_preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	0
correction_preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	0
correction_preflight	GPT-4o mini	1	0	0	0
correction_preflight	GPT-6 Luna Flex	1	0	0	0
preflight	Claude Haiku 5.5	1	0	0	0
preflight	Claude Sonnet 5.5	1	0	0	0
preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	0
preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	0
preflight	GPT-4o mini	1	0	0	0
preflight	GPT-6 Luna Flex	1	0	0	0

A failed HTTP attempt has no response, HTTP status 4xx/5xx, or an error payload inside HTTP 200. The count of attempts beyond one per sample measures additional HTTP attempts; it does not add new cases. A filter is a recorded response outcome and does not by itself trigger a replacement generation.

Tokens in final responses

Phase	Model	Input tokens	Output tokens	Reasoning tokens	Unknown input/output/reasoning samples
corpus	Claude Haiku 5.5	6894599	200731	0	352/352/352
corpus	Claude Sonnet 5.5	6996312	1265337	0	0/0/0
corpus	Gemini 3.1 Flash-Lite Flex	4601976	54039	0	0/0/0
corpus	Gemini 3.5 Flash-Lite Flex	4601976	54012	0	0/0/0
corpus	GPT-4o mini	5130216	57765	0	0/0/0
corpus	GPT-6 Luna Flex	5076216	270030	0	0/0/0
correction_preflight	Claude Haiku 5.5	88	4	0	0/0/0
correction_preflight	Claude Sonnet 5.5	89	4	0	0/0/0
correction_preflight	Gemini 3.1 Flash-Lite Flex	59	1	0	0/0/0
correction_preflight	Gemini 3.5 Flash-Lite Flex	59	1	0	0/0/0
correction_preflight	GPT-4o mini	68	1	0	0/0/0
correction_preflight	GPT-6 Luna Flex	67	5	0	0/0/0
preflight	Claude Haiku 5.5	96	4	0	0/0/0
preflight	Claude Sonnet 5.5	97	4	0	0/0/0
preflight	Gemini 3.1 Flash-Lite Flex	65	1	0	0/0/0
preflight	Gemini 3.5 Flash-Lite Flex	65	1	0	0/0/0
preflight	GPT-4o mini	74	1	0	0/0/0
preflight	GPT-6 Luna Flex	73	5	0	0/0/0

Token totals come from final output usage. They exclude usage missing from that field and do not count the usage of failed attempts as zero. Reported zero reasoning tokens do not prove the absence of unreported internal processing.

What conclusions are supported?

The results describe these six model/provider/reasoning conditions on this fixed corpus in one execution. Interpret agreement together with coverage and strict-format compliance. Review the condition tables before using the overall ranking: mHS has twice the weight of HS.

Models can differ on the same question, and the paired table quantifies those differences without changing the cases. Filters and unparsed responses describe limits in usable coverage. Explanatory text that is interpreted successfully can still violate the requested format.

Next, review source labels for cases with widespread disagreement and inspect mHS and reversed-order formulations. A logical audit should be a separate versioned analysis. Repeated independent runs would be needed to measure response variability. This report includes no confidence intervals, significance tests, causal claims, or cross-pilot comparisons.

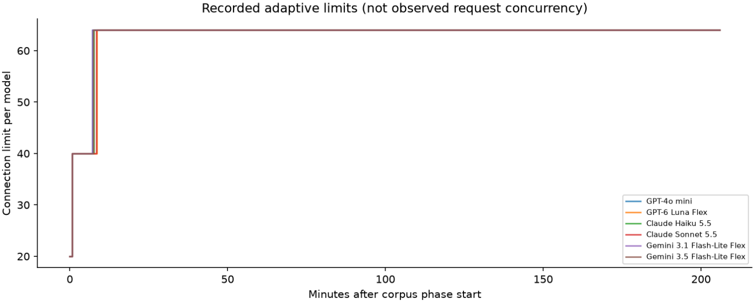
Operational appendix: original execution time and concurrency

Interval	Minutes
Preparation and dispatch after corpus start	0.72
First task start to last task completion	205.20
Finalization and exports after last task	15.61
Full corpus phase including closure	221.52

Corpus phase started at 2026-11-06 01:24:17 -03 and closed at 2026-11-06 05:05:49 -03. Task duration includes generation, waits, retries, metadata collection, and scoring. It is not pure provider processing time. The preflight is separate.

The run allowed six active tasks and 32 samples per task, for a 192-slot configured ceiling. Adaptive limits were min 10, start 20, max 64 per model. The controller can reduce limits after rate limits; transient errors pause growth. Limits are not measured request concurrency.

The observed peak was 192 concurrent samples and 140 concurrent captured HTTP intervals with both timestamps. Missing HTTP end timestamps prevent a complete bound on transport concurrency.



Model	Sample peak	HTTP interval peak	Mean sample seconds	P90 sample seconds	Mean HTTP seconds	P90 HTTP seconds
GPT-4o mini	32	32	6.54	9.99	1.19	3.85
GPT-6 Luna Flex	64	56	7.68	11.77	2.27	6.06
Claude Haiku 5.5	32	32	7.07	10.57	1.69	4.53
Claude Sonnet 5.5	64	53	7.75	11.33	2.46	5.34
Gemini 3.1 Flash-Lite Flex	32	32	6.47	9.85	1.12	3.70
Gemini 3.5 Flash-Lite Flex	32	32	6.78	10.26	1.41	4.12

Request and returned metadata checks

Phase	Model	Samples	Missing captured requests	Samples with condition faults	Samples with reasoning usage reported	Nonzero reasoning reported
corpus	Claude Haiku 5.5	54000	0	0	53648	0
corpus	Claude Sonnet 5.5	54000	0	0	54000	0
corpus	Gemini 3.1 Flash-Lite Flex	54000	0	0	54000	0
corpus	Gemini 3.5 Flash-Lite Flex	54000	0	0	54000	0
corpus	GPT-4o mini	54000	0	0	54000	0
corpus	GPT-6 Luna Flex	54000	0	0	54000	0
correction_preflight	Claude Haiku 5.5	1	0	0	1	0
correction_preflight	Claude Sonnet 5.5	1	0	0	1	0
correction_preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	1	0
correction_preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	1	0
correction_preflight	GPT-4o mini	1	0	0	1	0
correction_preflight	GPT-6 Luna Flex	1	0	0	1	0
preflight	Claude Haiku 5.5	1	0	0	1	0
preflight	Claude Sonnet 5.5	1	0	0	1	0
preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	1	0
preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	1	0
preflight	GPT-4o mini	1	0	0	1	0
preflight	GPT-6 Luna Flex	1	0	0	1	0

The returned routing table preserves reported provider and tier values, including empty sets. Missing evidence is not proof of a setting. The SQLite conditions table retains request-level fault descriptions. Native logs retain complete payloads and fields that this report does not project.

Evidence, definitions, and reproduction

Summary JSON · Validation record · Source and analysis hashes · SQLite analysis · Response table · Generation ledger · Cost conflicts

The report combines the original closed run with the verified correction and records both preflights separately. Inputs, original responses, native scores, logs, and exports are preserved. Every sample links to its native log and hash in the response table; every case includes source row, original identifiers, and record hashes in the dataset JSON files.

To rebuild offline, run uv run --locked transitivity report --run results/full/2026-11-06_01-23-40-040fd7b7 --output <new-directory>. The destination must not exist. No provider API requests are made. To browse the HTML bundle, serve the repository with uv run --locked python -m http.server 8000 and open this folder's index.html through localhost. All runtime assets and case shards are local.